Estimating Serial Intervals from Outbreak Symptom Onset Data

An Extension of the Vink et al. Transmission-Path Framework

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Table of contents

1 Introduction **1**

Notation **4**

2 Existing Method: Vink et al. **5**
2.1 The Original Normal Mixture Model . . . 5
2.1.1 Four latent transmission paths . . . 6
2.1.2 Discretisation of symptom-onset days . . . 8
2.2 EM algorithm . . . 11
2.3 Limitations . . . 12
2.3.1 Distributional assumption for underlying SI . . . 12
2.3.2 Number of transmission paths in the mixture model . . . 13
2.4 Motivation for the current project . . . 13

3 Methods **14**
3.1 Generalisation beyond the Normal distribution . . . 14
3.1.1 Ideal Log-Normal mixture framework . . . 14
3.1.2 Computational difficulty and lack of closed form . . . 15
3.1.3 Path-specific formulation under the Log-Normal extension . . . 16
3.1.4 Mixture model and discretised likelihood . . . 18
3.2 EM estimation for the Log-Normal extension . . . 19
3.3 Simulation study design . . . 21
3.3.1 Simulation scenarios . . . 21
3.3.2 Data generation and fitting protocol . . . 25
3.3.3 Evaluation metrics . . . 26
3.4 Data analysis . . . 26

4 Results **27**
4.1 Simulation . . . 27
4.1.1 Main simulation analysis (N = 100) . . . 27
4.1.2 Effect of sample size . . . 31
4.1.3 Investigation of anomalous SI estimates under low-dispersion scenario . . . 32
4.2 Sensitivity analysis . . . 34
4.2.1 Sensitivity to Initial Parameter Values . . . 34
4.3 Data analysis . . . 38

5 Software Implementation **40**
5.1 Package context and purpose . . . 40
5.2 Input layer: adding the Lognormal option . . . 40
5.3 Mixture-model layer: pathway-specific densities . . . 40
5.4 Estimation layer: adapting the EM algorithm . . . 41
5.5 Output layer . . . 42
5.6 Use in simulation and empirical analysis . . . 42

6 Discussion **42**
6.1 Limitations . . . 43
6.1.1 Restricted transmission-pathway structure . . . 44
6.1.2 Pathway overlap, initialization, and practical identifiability . . . 44
6.1.3 Distributional and observational assumptions . . . 45
6.2 Future work . . . 45

7 Appendix **46**
7.1 Appendix A. Additional Details of the EM Estimation Procedure . . . 46
7.1.1 A.1 Latent Pathway Responsibilities and the Vink Updating Principle . . . 46
7.1.2 A.2 PS-Weighted Updates Under the Log-Normal SI Model . . . 48
7.1.3 A.3 Treatment of Zero-Day ICC Intervals . . . 49
7.1.4 A.4 Initialization and Joint Convergence Criterion . . . 50
7.2 Appendix B. Assessment of Component-Density Approximations . . . 51
7.2.1 B.1 Co-primary pathway: Half-Normal approximation . . . 51
7.2.2 B.2 PT and PQ pathways: Distributional cost of the Fenton–Wilkinson approximation . . . 51
7.2.3 B.3 Numerical assessment of approximation accuracy . . . 52
7.3 Supplementary simulation results . . . 54

References **59**

1 Introduction

During the early phase of an outbreak, the most readily observable population-level signal is simply how the number of newly reported cases changes over time. While the speed at which these cases multiply tells us how rapidly the outbreak is expanding, this observed growth rate alone cannot uniquely determine the true transmissibility of the pathogen.

This is because epidemic growth reflects two distinct aspects of transmission. The first is the **magnitude of transmission**, commonly quantified by the reproduction number, R: the average number of secondary infections generated by an infected individual. At the beginning of an outbreak in a fully susceptible population, this quantity is referred to as the basic reproduction number, R_0. The second is the **timescale of transmission**, described by the generation time: the time between infection in a primary case (the infector) and infection in the corresponding secondary case (the infectee).

These two quantities jointly determine how quickly case numbers grow. For example, the same observed epidemic growth rate could arise from a pathogen that generates relatively many secondary infections over a longer generation time, or from one that generates fewer secondary infections but transmits them more rapidly. Consequently, translating epidemic growth into estimates of the reproduction number requires information about the timing of transmission.

In practice, however, the exact times at which individuals become infected are rarely observed. Generation times are therefore generally not directly measurable. Symptom-onset times are much more readily recorded, making the **serial interval (SI)** a practical observable proxy. The SI is defined as the time between symptom onset in a primary case (the infector) and symptom onset in the corresponding secondary case (the infectee). Although the SI is not identical to the generation time, it is commonly used to inform the transmission-timescale distribution required to estimate reproduction numbers from epidemic growth[1,2]. Accurate estimation of the SI is therefore important for quantifying disease transmissibility.

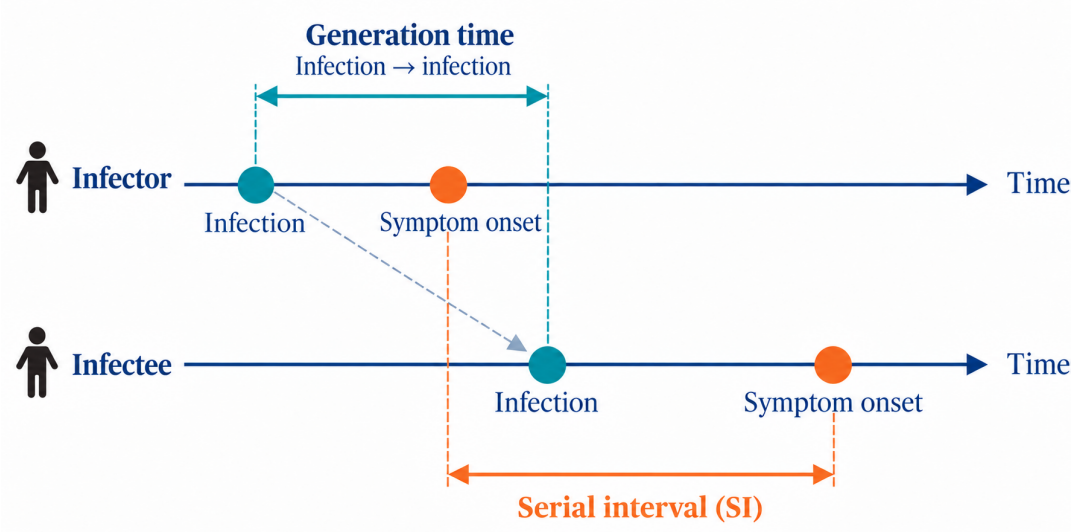


Figure 1: Generation time and serial interval for one infector–infectee pair. The horizontal arrows show time rather than a scaled numerical axis; the upper and lower timelines represent the infector and infectee. Teal markers denote infection and orange markers denote symptom onset. The teal span measures infection-to-infection generation time, whereas the orange span measures onset-to-onset serial interval.

Despite its epidemiological importance, the SI is often difficult to measure directly in real outbreak settings. Surveillance data commonly record symptom-onset dates, but the transmission links connecting individual cases are typically unobserved. In other words, researchers may know when each case developed symptoms without knowing exactly who infected whom. As illustrated in panel A of Figure 2, several different transmission networks may therefore be compatible with the same observed onset data.

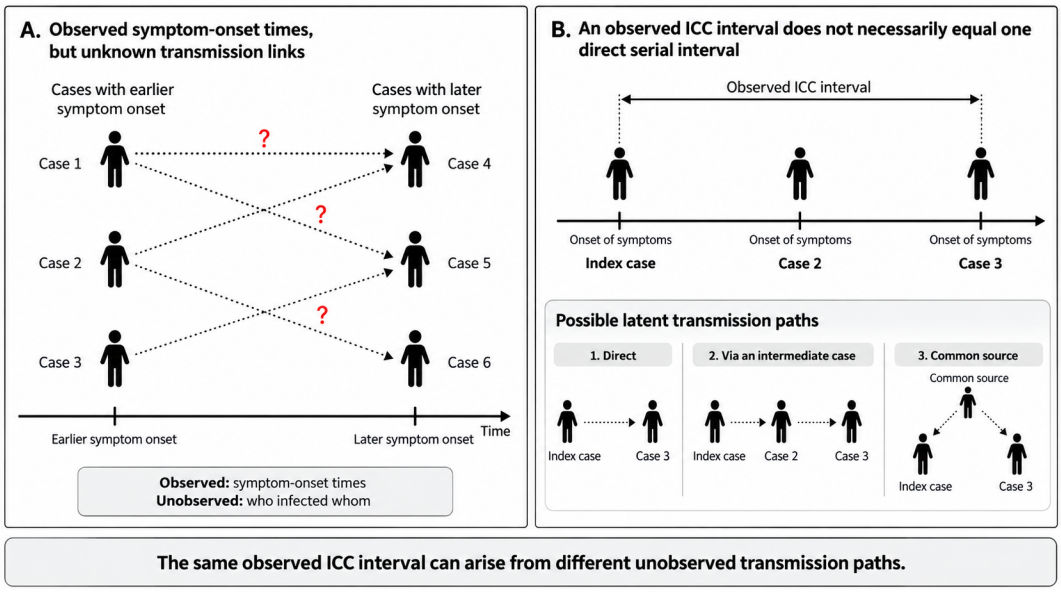


Figure 2: Why onset dates alone do not identify transmission links. Panel A orders cases from earlier to later symptom onset along a schematic time axis; dotted arrows mark possible but unobserved infector–infectee links. Panel B measures the ICC interval from the index case to Case 3 and contrasts direct, intermediate-case, and common-source pathways that could produce it. There is no numerical vertical axis.

In household or small-cluster settings, the underlying transmission network is typically unobserved. However, one well-defined quantity can still be constructed directly from the available symptom-onset data: the index case-to-case (ICC) interval. By identifying the first recognised symptomatic individual as the index case, the ICC interval for each subsequent case is defined as the elapsed time between that case's symptom onset and the symptom onset of the index case. For a non-index case i, let S_i be that case's symptom-onset time and S_0 the index case's symptom-onset time. The underlying continuous ICC interval is

$$C_i = S_i - S_0.$$

Here, C_i is the continuous ICC interval for non-index case i; S_i and S_0 are the symptom-onset times of case i and the index case, respectively.

Because the index case is selected as the first symptomatic case, an ICC interval is non-negative. This differs from a direct SI, which can be negative when the infectee develops symptoms before the infector. Importantly, calculating an ICC interval requires only observed symptom-onset times and does not require the underlying infector–infectee pair to be known. It therefore provides a practical way to summarise the timing information available from household or small-cluster outbreak data.

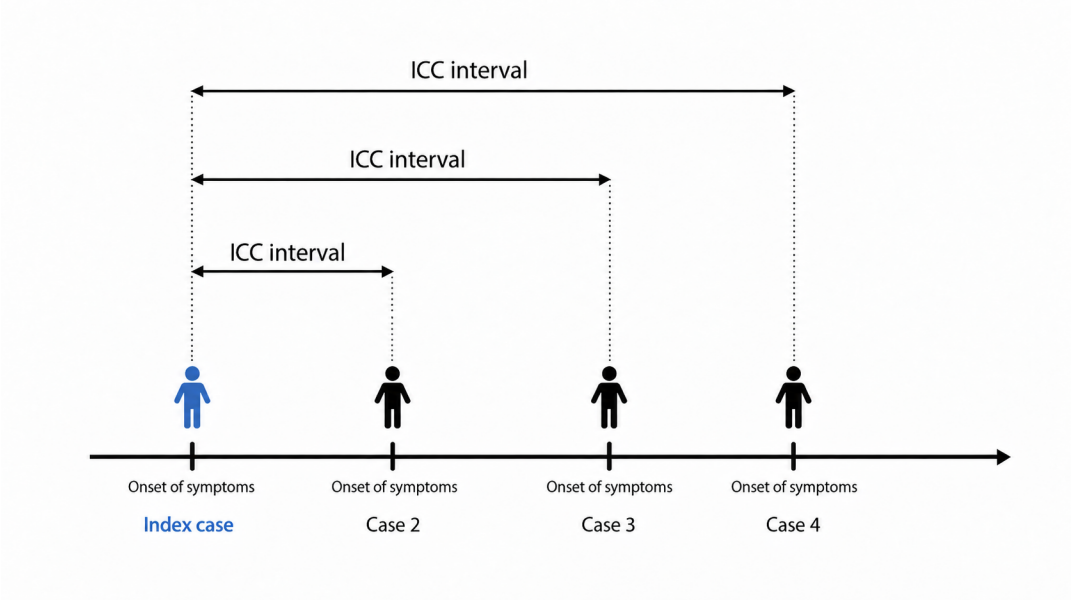


Figure 3: Construction of index case-to-case (ICC) intervals. The horizontal timeline orders symptom onsets; each arrow starts at the index case's onset and ends at a later case's onset. The three arrows illustrate that a separate ICC interval is observed for each subsequent case without identifying who infected whom. This is a schematic, so neither axis has a numerical scale.

However, an ICC interval is not necessarily a direct serial interval. The ICC is defined relative to the index case, whereas the SI is defined specifically between a true infector and their infectee. Because the underlying transmission chain is unobserved, the same ICC interval may be compatible with several different transmission pathways, including direct transmission, transmission through intermediate cases, or infection from a common source, as illustrated in panel B of Figure 2. Each pathway implies a different relationship between the ICC interval and the underlying SI.

Consequently, symptom-onset data alone cannot determine which transmission pathway generated a particular ICC observation. Estimating the underlying SI distribution requires a statistical framework that accounts for uncertainty in these latent transmission pathways. The following section introduces the mixture-model framework proposed by Vink et al. (2014) to address this problem.

Notation

The main notation used throughout this report is summarised in Table 1. Additional symbols specific to individual derivations are defined where they first appear.

Table 1: Main notation used in the transmission-pathway mixture framework.

Symbol	Description
X, X_k	Underlying direct serial interval (SI) and its independent copies
S_i, S_0	Symptom-onset times of case i and the index case
C_i	Continuous index case-to-case (ICC) interval for observation i
T_i, t_i	Observed integer-day ICC interval and its realised value
μ, σ^2	Mean and variance of the underlying SI on the natural scale
m, s	Natural-scale SI mean and standard deviation, particularly in simulations and estimation
η, ζ^2	Mean and variance of the log-transformed SI
$\mathcal{G}$	Set of latent transmission pathways: CP, PS, PT, and PQ
Z_i	Unobserved transmission pathway associated with observation i
$w_j, \mathbf{w}$	Mixture weight for pathway j and the vector of pathway weights
$f_j(c)$	Continuous ICC probability density under pathway j
p_{ij}	Discretised probability of ICC observation i under pathway j
$\tau_{ij}^{(r)}$	Posterior responsibility of pathway j for observation i at EM iteration r
N	Number of ICC intervals in a dataset

2 Existing Method: Vink et al.

Because the transmission pathway associated with each ICC interval is unknown, an empirical ICC dataset may combine intervals generated through different numbers or configurations of transmission steps. Vink et al. (2014) addressed this ambiguity by modelling the ICC distribution as a finite mixture whose components correspond to four epidemiologically interpretable latent transmission pathways[3]. This construction links the ICC intervals to the underlying SI distribution while accounting for uncertainty about who infected whom. The following methodology describes how these four pathway-specific components are constructed and combined.

2.1 The Original Normal Mixture Model

In the model of Vink et al., let X denote an underlying serial interval, defined as the time between symptom onset in a primary case and symptom onset in the corresponding secondary case. The serial

interval is assumed to follow a Normal distribution,

$$X \sim N(\mu, \sigma^2),$$

where $N(a, b)$ denotes a Normal distribution with mean a and variance b, the symbol $\sim$ means "is distributed as", μ is the mean serial interval, and σ^2 is the variance of the serial interval.

When several serial intervals appear in one pathway, X_k denotes the k-th independent draw from this distribution, with $k = 1, 2, 3, ...$; thus X_1, X_2, and X_3 are independent and identically distributed copies of X.

The Normal distribution provides both epidemiological flexibility and mathematical convenience. Its support over the entire real line allows negative serial intervals, which can occur when an infectee develops symptoms before their infector. In addition, sums and differences of independent Normal variables remain Normal, giving closed-form expressions for the path-specific distributions and avoiding the need for numerical integration.

2.1.1 Four latent transmission paths

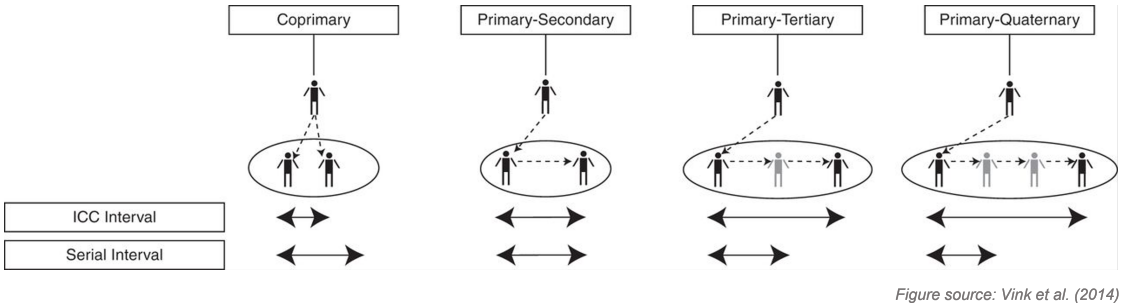


Figure 4: Four latent pathways in the Vink transmission model (adapted from[3]). The columns show coprimary (CP), primary-secondary (PS), primary-tertiary (PT), and primary-quaternary (PQ) transmission; dashed arrows indicate infection links and grey figures indicate intermediate cases. Within each column, the upper horizontal arrow spans the ICC interval and the lower arrow spans one underlying direct serial interval. Arrow lengths are schematic rather than quantitative.

Let

$$\mathcal{G} = \{\text{CP}, \text{PS}, \text{PT}, \text{PQ}\}$$

denote the set of four possible latent transmission paths, and let $Z_i \in \mathcal{G}$ denote the unobserved path associated with observation i. Here, i indexes ICC observations and $j \in \{1, 2, 3, 4\}$ indexes CP,

PS, PT, and PQ, respectively. The symbol f_j denotes the continuous ICC density under pathway j. Each pathway represents a distinct transmission history and therefore implies a different mathematical relationship between the ICC interval and the underlying serial intervals. Because ICC intervals are non-negative by definition, the path-specific ICC quantities are expressed as absolute values of the relevant sums or differences of Normal serial intervals:

- **Co-primary transmission (f_1):** The index case and the other observed case are independently infected by the same person outside the household. The ICC interval is the absolute difference between two independent serial intervals from this common source,

$$C_{\text{CP}} = |X_1 - X_2|, \qquad X_1 - X_2 \sim N(0, 2\sigma^2).$$

- **Primary-secondary transmission (f_2):** The index case directly infects the other observed case. The ICC interval therefore arises from one direct SI,

$$C_{\text{PS}} = |X_1|, \qquad X_1 \sim N(\mu, \sigma^2).$$

- **Primary-tertiary transmission (f_3):** The index case infects an intermediate case, which then infects the other observed case. The ICC interval spans two successive serial intervals,

$$C_{\text{PT}} = |X_1 + X_2|, \qquad X_1 + X_2 \sim N(2\mu, 2\sigma^2).$$

- **Primary-quaternary transmission (f_4):** Transmission from the index case to the other observed case occurs through two intermediate cases. The ICC interval spans three successive serial intervals,

$$C_{\text{PQ}} = |X_1 + X_2 + X_3|, \qquad X_1 + X_2 + X_3 \sim N(3\mu, 3\sigma^2).$$

Because the transmission path responsible for an individual ICC observation is unobserved, the overall ICC distribution is represented as a finite mixture of these four path-specific components. Let $c \geq 0$ denote a possible value of the continuous ICC interval, and let w_j denote the mixture weight, or population proportion, for component f_j. Write $\mathbf{w} = (w_1, w_2, w_3, w_4)$ for the full vector of pathway weights. The weights satisfy

$$w_j \geq 0, \qquad \sum_{j=1}^{4} w_j = 1,$$

the overall continuous ICC density is formulated as the finite mixture

$$f(c \mid \mu, \sigma^2, \mathbf{w}) = w_1 f_1(c \mid \sigma^2) + w_2 f_2(c \mid \mu, \sigma^2) + w_3 f_3(c \mid \mu, \sigma^2) + w_4 f_4(c \mid \mu, \sigma^2).$$

All four mixture components are derived from the same underlying SI distribution and therefore share the common parameters μ and σ^2. The mixture weights represent the relative contribution of the different transmission pathways to the ICC data.

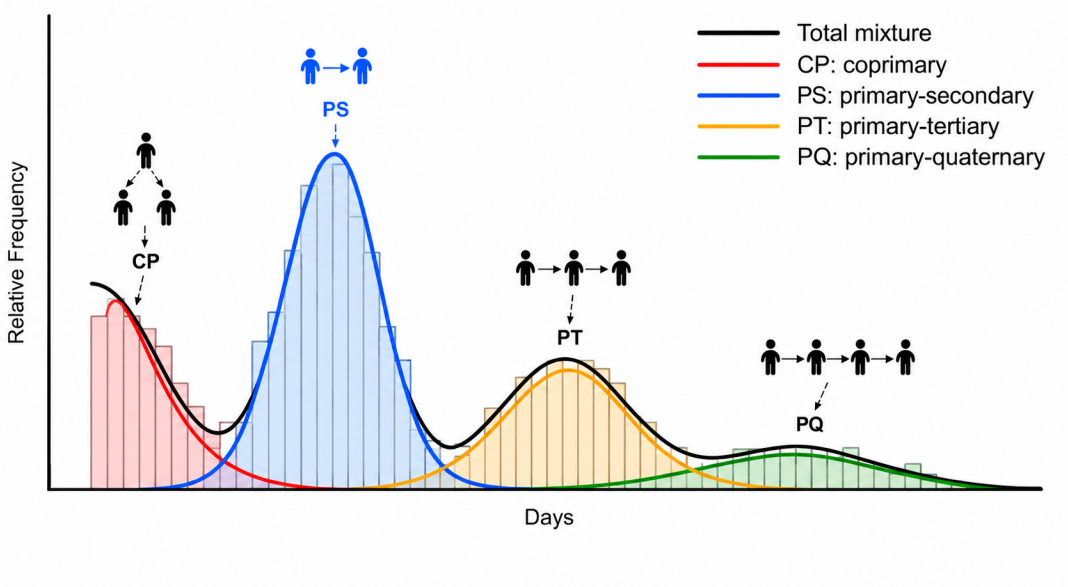


Figure 5: Schematic four-component ICC mixture under the Normal SI model. The horizontal axis shows ICC days and the vertical axis shows relative frequency; coloured curves and bars depict co-primary (CP, red), primary-secondary (PS, blue), primary-tertiary (PT, orange), and primary-quaternary (PQ, green) contributions. The black curve is their combined mixture. The schematic highlights how pathway-specific timing can create multiple modes; its axis values are illustrative.

2.1.2 Discretisation of symptom-onset days

The mixture model above describes the ICC interval on a continuous time scale, whereas the data available in practice are usually recorded only as integer numbers of days. The purpose of this discretisation step is therefore to connect the continuous ICC model to the observed data by deriving the probability of observing an ICC interval of exactly t days.

Let C_i denote the underlying continuous ICC interval for observation i, and let T_i denote its observed integer-day counterpart, with observed value t_i. For example, if the index case is recorded as developing symptoms on 1 May and a later case on 4 May, then the recorded ICC interval is $T_i = 3$ days. However, the exact onset time of either case within 1 May or 4 May is unknown. Thus, both endpoints of the interval are only known to lie somewhere within their respective 24-hour periods. This form of observation is commonly referred to as **doubly interval-censored data**[3].

Following Vink et al.[3], the exact onset time is assumed to be uniformly distributed within the reported day. Let D_{1i} and D_{2i} denote the recorded onset days of the index case and the later case. Their exact onset times can be written as $S_{1i} = D_{1i} + U_{1i}$ and $S_{2i} = D_{2i} + U_{2i}$, where $U_{1i}, U_{2i} \overset{\text{iid}}{\sim} \text{Uniform}(0, 1)$ represent the unknown within-day onset times. The subscripts 1 and 2 identify the index and later case in pair i, and "iid" means independent and identically distributed.

The recorded integer-day difference is $T_i = D_{2i} - D_{1i}$, whereas the true continuous ICC interval is

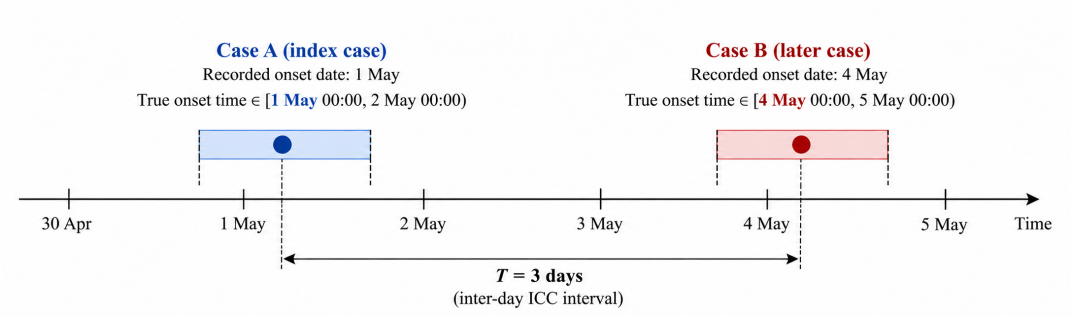


Figure 6: Doubly interval-censored onset dates. The horizontal axis is calendar time; blue and red boxes show the 24-hour windows for the index case and later case, and dots illustrate their unknown exact onset times. Although the recorded ICC interval is 3 days between 1 and 4 May, both true onset times vary within their respective day-long windows.

$$C_i = S_{2i} - S_{1i} = T_i + (U_{2i} - U_{1i}).$$

Let $W_i = U_{2i} - U_{1i}$ be the within-day timing difference for observation i. Because it is the difference between two independent $\text{Uniform}(0, 1)$ variables, W_i follows the symmetric triangular density g; the argument w below denotes a possible value of W_i:

$$g(w) = \begin{cases} 1 - |w|, & |w| \le 1, \\ 0, & |w| > 1. \end{cases}$$

The density is highest at $w = 0$, corresponding to the two cases developing symptoms at approximately the same time of day within their respective recorded days. It decreases linearly towards zero as w approaches -1 or 1, which would require one onset to occur near the beginning of a day and the other near the end.

Therefore, if the observed interval is $T_i = t_i$, the corresponding continuous interval C_i may lie between $t_i - 1$ and $t_i + 1$. For example, an ICC interval of 3 days is compatible with continuous intervals between 2 and 4 days.

To calculate the probability of observing exactly t_i days under the continuous model, the triangular density is shifted to be centred at t_i. Define

$$g_{t_i}(c) = \begin{cases} 1 - |c - t_i|, & |c - t_i| \leq 1, \\ 0, & |c - t_i| > 1, \end{cases}$$

where $g_{t_i}(c)$ gives the weight associated with a possible continuous ICC value c when the recorded interval is t_i days.

Let $f(c \mid \mu, \sigma^2, \mathbf{w})$ denote the continuous ICC mixture density derived in the previous section. The required discrete probability is obtained by integrating this continuous density over all values of c compatible with the observed integer interval, weighted by the triangular kernel. In the integral, dc indicates integration with respect to the continuous ICC value c:

$$P(T_i = t_i \mid \mu, \sigma^2, \mathbf{w}) = \int_{\max(0, t_i - 1)}^{t_i + 1} f(c \mid \mu, \sigma^2, \mathbf{w}) g_{t_i}(c)\, dc.$$

The lower limit is truncated at zero because ICC intervals have non-negative support. This equation provides the key link between the continuous model and the discrete data: it converts the continuous ICC density into the probability of observing an integer-day interval of exactly t_i days.

Since the continuous ICC density is itself a mixture,

$$f(c \mid \mu, \sigma^2, \mathbf{w}) = \sum_{j=1}^{4} w_j f_j(c \mid \mu, \sigma^2),$$

the same discretisation can be applied separately to each transmission-path component. Define

$$p_{ij}(\mu, \sigma^2) = \int_{\max(0, t_i - 1)}^{t_i + 1} f_j(c \mid \mu, \sigma^2) g_{t_i}(c)\, dc$$

as the probability of observing t_i days if observation i arose from transmission pathway j; thus the first subscript identifies the observation and the second identifies the pathway. By linearity of integration, the overall probability can then be written as

$$P(T_i = t_i \mid \mu, \sigma^2, \mathbf{w}) = \sum_{j=1}^{4} w_j p_{ij}(\mu, \sigma^2).$$

Thus, each ICC interval is linked to the continuous model through four path-specific discrete probabilities, which are then combined using the mixture weights.

Once a model-based probability has been obtained for every observed interval, these probabilities can be combined across the full dataset. Let $\mathbf{t} = (t_1, \ldots, t_n)$ denote the n ICC intervals. Assuming that the observations are independent, their joint probability under a given set of model parameters is the product of the individual probabilities. When viewed as a function of the unknown parameters with the observed data held fixed, this joint probability defines the likelihood:

$$L(\mu, \sigma^2, \mathbf{w}; \mathbf{t}) = \prod_{i=1}^{n} \left\{ \sum_{j=1}^{4} w_j p_{ij}(\mu, \sigma^2) \right\}.$$

Here, L denotes the observed-data likelihood, $\mathbf{t}$ is the data vector, n is the number of ICC observations, and $\prod$ denotes multiplication over observations. The likelihood therefore measures how well a proposed set of SI parameters and mixture weights explains the observed integer-day ICC data. Parameter estimation seeks values that maximise this likelihood. Because the transmission pathway associated with each observation remains unobserved, direct maximisation is naturally formulated as a latent-variable problem, motivating the expectation–maximisation algorithm described in the next section.

2.2 EM algorithm

Because the true transmission route underlying each ICC observation is not observed, estimation is naturally formulated as a latent-variable problem. Vink et al. addressed this using an expectation–maximisation (EM) algorithm. At each iteration, the E-step computes the posterior pathway responsibilities for every ICC observation, and the M-step updates the mixture weights and SI parameters accordingly. A schematic overview is shown in Figure 7; the detailed updating equations are given in the Appendix.

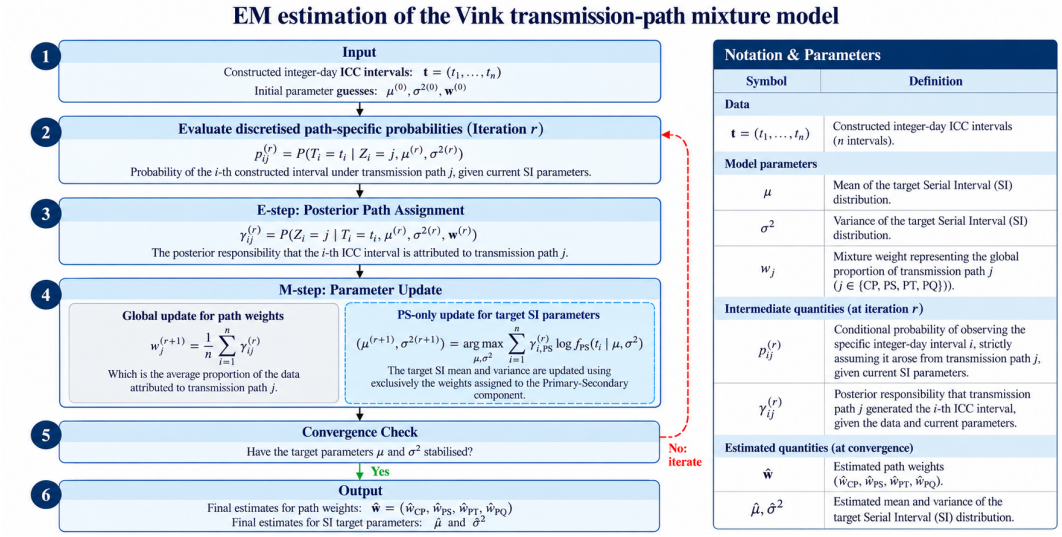


Figure 7: EM workflow for the original Vink transmission-path mixture model. The diagram is a process flow, with no numerical axes: observed integer-day ICC intervals and initial parameters enter at the top; path-specific probabilities feed the E-step responsibilities; the M-step updates all four pathway weights but updates SI mean and variance using PS responsibilities only. The loop repeats until the parameter-change criterion is met.

2.3 Limitations

While the Vink framework provides a practical approach for estimating SI parameters without known transmission pairs, its performance depends on several modelling assumptions. Two limitations are particularly relevant here: the assumed distributional form of the underlying SI and the restricted number of transmission paths represented in the mixture model.

2.3.1 Distributional assumption for underlying SI

A primary limitation of the Vink framework is its reliance on a prespecified distribution for the underlying serial interval (SI). In the 2014 formulation, the SI is assumed to follow a symmetric Normal distribution, which conveniently allows negative SI values when an infectee develops symptoms before their infector. The non-negative ICC intervals are then constructed from absolute sums or differences of these underlying SI variables and subsequently discretised for symptom-onset dates[3].

The Normal assumption is mathematically convenient, but its symmetry may become restrictive when the true SI distribution is asymmetric or has a pronounced upper tail. Serial-interval shape is not universal across infectious diseases. For example, the directly reconstructed serial intervals of primary pneumonic plague were strongly positively skewed, with a reported skewness of 2.3 and a Gamma distribution used to describe their shape[4]. In an early analysis of COVID-19 infector–infectee pairs, a

Log-Normal distribution provided the best-fitting serial-interval model among the candidate distributions considered[5]. These examples demonstrate that right-skewed SI distributions can arise in practice, although they do not imply that all serial intervals are right-skewed.

The choice of distribution has also varied within applications of the Vink framework itself. Vink's earlier thesis assumed a Gamma SI distribution because of its flexible shape[6], whereas the 2014 analysis used a Normal distribution as the primary specification and repeated the analysis under a Gamma assumption as a sensitivity analysis, obtaining broadly similar results[3]. More recently, Ainslie et al. applied the framework to scabies outbreaks using a Normal SI distribution and found that a Gamma sensitivity model provided a poorer fit to those data[7]. Thus, there is no empirical basis for treating a right-skewed distribution as universally preferable to the Normal assumption.

Taken together, these findings suggest that the distributional form of the underlying SI should be regarded as a modelling assumption rather than a fixed feature of the Vink framework. While the Normal specification is convenient and performs adequately in some settings, other distributions may be more appropriate when the SI is asymmetric, strongly right-skewed, or otherwise departs from Normality. The suitability of alternative distributional assumptions therefore remains an open methodological question and warrants further investigation.

2.3.2 Number of transmission paths in the mixture model

A further limitation concerns the restricted set of transmission pathways represented in the mixture. Vink et al. considered four components—co-primary, primary-secondary, primary-tertiary, and primary-quaternary—and excluded longer transmission chains on the grounds that they were unlikely in household settings[3]. This is a reasonable simplification for small households or similarly contained outbreaks, for which the framework was originally developed. However, the four-path structure may be less adequate in larger clusters or more complex transmission settings, where additional generations of transmission or more complicated transmission histories may occur. The applicability of the model is therefore partly determined by whether these four pathways provide a reasonable approximation to the transmission structure of the outbreak being analysed.

2.4 Motivation for the current project

The Vink framework provides a practical way to infer the underlying serial interval (SI) distribution from constructed ICC intervals by modelling them as a mixture of plausible transmission paths. Existing implementations have primarily relied on Normal and Gamma assumptions for the underlying SI[3,7]. These distributions provide convenient and useful starting points, but no single distributional form is expected to describe transmission timing equally well across all infectious diseases. As discussed above, empirical SI distributions can vary substantially in shape, and positively skewed serial intervals have been reported in some settings[4,5]. This motivates exploring an additional right-skewed

distributional assumption. The Log-Normal distribution was selected as a natural starting point to investigate whether it can provide improved fit or parameter recovery when the underlying SI is positive and right-skewed.

A second motivation is to broaden the practical and methodological flexibility of the existing implementation R package. The `{mitey}` package provides an implementation of the Vink approach, with the current SI modelling framework centred on Normal and Gamma specifications. Extending it to include a Log-Normal option would give researchers a wider choice of distributional assumptions when analysing diseases with different transmission-time characteristics. More broadly, the Log-Normal case provides a useful proof of concept for extending the transmission-path framework beyond distributions for which the path-specific components are analytically convenient. Developing a workable estimation strategy in this setting may therefore provide a foundation for future extensions to other plausible SI distributions, including models that require numerical integration or optimisation.

To address these aims, the following Methods chapter first reconstructs the ideal lognormal mixture framework, then introduces tractable approximations for its non-closed-form components and develops a robust, customized EM procedure to resolve the resulting numerical singularities. The chapter will finally concludes by showing how empirical data and a simulation study were used to test plausibility of the method from parameter-recovery performance.

3 Methods

3.1 Generalisation beyond the Normal distribution

The present extension therefore retains the same four latent transmission paths, mixture weights, ICC construction, and triangular discretisation model introduced in the previous chapter, while changing only the assumed distribution of the underlying SI.

To maintain consistency with the preceding notation, μ and σ^2 continue to denote the mean and variance of the underlying SI on the original time scale. Likewise, the four transmission-path component densities remain denoted by $f_1, \dots, f_4$, with mixture weights $w_1, \dots, w_4$, and p_{ij} continues to denote the discretised probability of the i-th ICC interval under transmission path j.

3.1.1 Ideal Log-Normal mixture framework

Let X_k denote the k-th successive direct serial interval. Under the proposed extension,

$$X_k \overset{\text{iid}}{\sim} \text{Lognormal}(\mu, \sigma^2), \qquad k = 1, 2, 3, \dots,$$

where $k = 1, 2, 3, ...$, "iid" means independent and identically distributed, and μ and σ^2 are the mean and variance of X_k on the original time scale. In the notation Lognormal(μ, σ^2) used here, the two arguments therefore describe natural-scale moments rather than the mean and variance of $\log X_k$. Equivalently, the corresponding log-scale distribution is

$$\log X_k \sim N\left(\log(\mu) - \frac{1}{2}\log\left(1 + \frac{\sigma^2}{\mu^2}\right),\ \log\left(1 + \frac{\sigma^2}{\mu^2}\right)\right).$$

Let $f_X(x \mid \mu, \sigma^2)$ denote the resulting Log-Normal density of a direct SI evaluated at a possible duration $x > 0$. The four transmission paths retain exactly the same epidemiological interpretation as before, but their path-specific distributions now arise from independent Log-Normal serial intervals:

$$C_{\text{CP}} = |X_1 - X_2|, \qquad C_{\text{PS}} = X_1, \qquad C_{\text{PT}} = X_1 + X_2, \qquad C_{\text{PQ}} = X_1 + X_2 + X_3.$$

Because the Log-Normal distribution has positive support, the PS, PT, and PQ intervals are naturally non-negative and no folding is required for these three components. The absolute value remains necessary for the CP path because its ICC interval is defined by the difference between two independent serial intervals.

3.1.2 Computational difficulty and lack of closed form

The main computational challenge of the Log-Normal extension arises from the loss of the analytical closure available under the Normal assumption. While sums and differences of independent Normal variables remain Normal, the corresponding operations for Log-Normal variables generally do not lead to distributions with convenient closed-form densities. In particular, the CP component involves the absolute difference of two independent Log-Normal variables, whereas the PT and PQ components involve sums of two or three Log-Normal variables. Their exact densities therefore require numerical integration or convolution.

This becomes important for estimation because the path-specific densities must subsequently be discretised through the triangular observation model and evaluated repeatedly throughout the likelihood and EM algorithm. Direct use of the exact Log-Normal expressions would therefore introduce substantially greater computational cost than in the original Normal framework.

The following sections consider each transmission path separately and describe the computational formulation used to obtain tractable path-specific component densities while preserving the underlying transmission-path interpretation.

3.1.3 Path-specific formulation under the Log-Normal extension

The four transmission paths retain the same epidemiological interpretation as in the original Vink framework. However, changing the underlying SI distribution from Normal to Log-Normal alters the mathematical form of several path-specific components. Table 2 summarises the ideal construction and the working formulation used for each transmission path.

Table 2: Path-specific formulation under the Log-Normal SI assumption.

Transmission path	Ideal construction	Approximation required?	Working formulation
Co-primary (CP)	$C_{\text{CP}} = \lvert X_1 - X_2 \rvert$	Yes	Half-Normal approximation
Primary-secondary (PS)	$C_{\text{PS}} = X_1$	No	Exact Log-Normal
Primary-tertiary (PT)	$C_{\text{PT}} = X_1 + X_2$	Yes	Fenton–Wilkinson approximation
Primary-quaternary (PQ)	$C_{\text{PQ}} = X_1 + X_2 + X_3$	Yes	Fenton–Wilkinson approximation

The PS component remains available in closed form because it consists of a single Log-Normal SI. In contrast, the CP component involves a difference of Log-Normal variables, while PT and PQ involve sums of Log-Normal variables; these do not have convenient closed-form densities. Computationally tractable approximations are therefore introduced for these three components.

3.1.3.1 Co-primary transmission: Half-Normal approximation

Under co-primary transmission, the index case and the other case are independently infected by the same external source. The ideal ICC interval is therefore $C_{\text{CP}} = |X_1 - X_2|$, where X_1 and X_2 are independent draws from the same underlying Log-Normal SI distribution.

The absolute difference of two independent Log-Normal variables does not have a convenient closed-form density. Direct evaluation would therefore require repeated numerical integration during likelihood evaluation and EM estimation. To avoid this computational burden, the signed difference $X_1 - X_2$ is approximated by a zero-centred Normal distribution with the same variance. Since

$$\text{Var}(X_1 - X_2) = 2\sigma^2,$$

the working approximation is

$$C_{\text{CP}} \approx \left| N(0, 2\sigma^2) \right|,$$

which corresponds to a Half-Normal distribution with scale $\sqrt{2}\sigma$. Its density is

$$f_1(c \mid \sigma^2) = \frac{1}{\sigma\sqrt{\pi}} \exp\left(-\frac{c^2}{4\sigma^2}\right), \qquad c \geq 0.$$

Here, $\approx$ denotes a working approximation rather than equality, and $\exp(\cdot)$ is the exponential function. This approximation retains the variance scale of the signed difference before folding and naturally places substantial density near zero, which is consistent with co-primary cases having closely spaced symptom-onset times. However, it is not the exact distribution of $|X_1 - X_2|$, and its shape depends only on σ^2, whereas the exact Log-Normal difference distribution depends on both μ and σ^2.

3.1.3.2 Primary-secondary transmission: Exact Log-Normal component

Under primary-secondary transmission, the index case directly infects the other case, so the ICC interval consists of a single direct serial interval: $C_{\text{PS}} = X_1$.

No approximation is required for this component. Under the assumed Log-Normal SI model,

$$f_2(c \mid \mu, \sigma^2) = f_X(c \mid \mu, \sigma^2), \qquad c > 0.$$

The PS component therefore remains the exact target SI distribution. As in the original Vink framework, it is this component that provides the direct information used to estimate the SI mean μ and variance σ^2.

3.1.3.3 Primary-tertiary transmission: Fenton–Wilkinson approximation

Under primary-tertiary transmission, the ICC interval contains two successive direct serial intervals, so $C_{\text{PT}} = X_1 + X_2$. Its exact natural-scale mean and variance are 2μ and $2\sigma^2$, respectively.

Although these moments are available directly, the sum of two independent Log-Normal variables does not have a convenient closed-form density. The PT component is therefore approximated using the Fenton–Wilkinson method, which replaces the exact sum with a single Log-Normal distribution having the same first two natural-scale moments:

$$C_{\text{PT}} \approx \text{Lognormal}(2\mu, 2\sigma^2).$$

The Fenton–Wilkinson approximation therefore preserves the exact mean and variance of the two-step transmission interval while providing a density that can be evaluated efficiently during likelihood calculation and EM estimation.

3.1.3.4 Primary-quaternary transmission: Fenton–Wilkinson approximation

Under primary-quaternary transmission, the ICC interval spans three successive direct serial intervals: $C_{\text{PQ}} = X_1 + X_2 + X_3$. The corresponding exact natural-scale mean and variance are 3μ and $3\sigma^2$.

As with the PT component, the exact sum of three independent Log-Normal variables does not have a convenient closed-form density. The same Fenton–Wilkinson moment-matching strategy is therefore applied:

$$C_{\text{PQ}} \approx \text{Lognormal}(3\mu, 3\sigma^2).$$

This approximation again preserves the exact first two moments of the multi-generation interval while avoiding repeated numerical convolution. Together, the Half-Normal approximation for CP, the exact Log-Normal representation for PS, and the Fenton–Wilkinson approximations for PT and PQ provide the four working component densities used in the subsequent mixture likelihood and EM estimation.

The distributional accuracy of the proposed working approximations is assessed numerically in Figure 15, with further details provided in Appendix B.

3.1.4 Mixture model and discretised likelihood

After defining the four path-specific working components, the overall mixture retains exactly the same structure as in the original framework:

$$f(c \mid \mu, \sigma^2, \mathbf{w}) = \sum_{j=1}^{4} w_j f_j(c \mid \mu, \sigma^2),$$

with

$$w_j \geq 0, \qquad \sum_{j=1}^{4} w_j = 1.$$

The triangular discretisation model is also unchanged. For the i-th constructed ICC interval $T_i = t_i$, the discrete probability under transmission path j is

$$p_{ij}(\mu, \sigma^2) = \int_{\max(0, t_i - 1)}^{t_i + 1} f_j(c \mid \mu, \sigma^2) g_{t_i}(c)\, dc.$$

The corresponding mixture probability is

$$P(T_i = t_i \mid \mu, \sigma^2, \mathbf{w}) = \sum_{j=1}^{4} w_j p_{ij}(\mu, \sigma^2),$$

and the likelihood for the complete set of constructed ICC intervals $\mathbf{t} = (t_1, \ldots, t_n)$ is

$$L(\mu, \sigma^2, \mathbf{w}; \mathbf{t}) = \prod_{i=1}^{n} \left\{ \sum_{j=1}^{4} w_j p_{ij}(\mu, \sigma^2) \right\}.$$

The transmission-path interpretation, mixture structure, discretisation model, and target parameters are therefore unchanged from the original Vink framework. The principal modification lies in the path-specific component densities: PS remains exact, whereas CP, PT, and PQ use computationally tractable approximations to the corresponding Log-Normal difference or sum distributions.

3.2 EM estimation for the Log-Normal extension

The Log-Normal extension retains the overall EM structure of the original Vink framework: membership of true transmission path is treated as latent, the E-step evaluates the posterior responsibility of each transmission path for every constructed ICC interval, and the M-step updates the pathway weights and the parameters of the target SI distribution. However, several aspects of the estimation procedure require adaptation to accommodate the Log-Normal SI assumption and the working path-specific components introduced above.

Figure 8 provides an overview of the resulting EM algorithm. At each iteration, the discretised probabilities of the four transmission paths are evaluated using the current SI parameters and pathway weights. These probabilities are combined in the E-step to obtain posterior responsibilities. The mixture weights are then updated using information from all four transmission paths, whereas the target SI parameters are updated using the responsibility assigned to the primary-secondary component, following the pathway-specific updating principle of the original Vink algorithm.

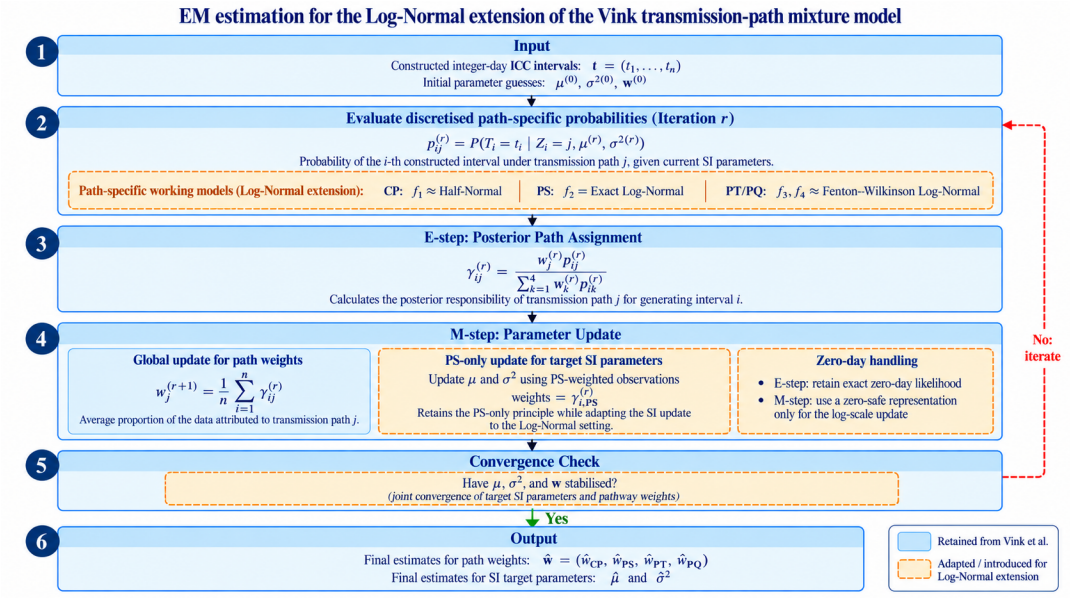


Figure 8: EM workflow for the Log-Normal extension. The top-to-bottom flow has no numerical axes: CP uses a Half-Normal working approximation, PS an exact Log-Normal component, and PT/PQ Fenton–Wilkinson approximations. The E-step assigns pathway responsibilities; the M-step updates four weights and PS-based SI parameters while handling zero-day observations; the convergence check includes both SI parameters and weights. Blue boxes retain the Vink structure and orange outlines mark adaptations.

Initialization and Path-Specific Evaluation. The algorithm initializes with the constructed integer-day ICC intervals alongside initial guesses for the SI parameters and mixture weights. To accommodate the Log-Normal extension, the evaluation of path-specific components is adapted: the primary-secondary (PS) component is evaluated exactly, while the co-primary (CP) component employs a Half-Normal approximation, and the primary-tertiary (PT) and primary-quaternary (PQ) components rely on Fenton–Wilkinson approximations.

E-step: Posterior Path Assignment. The expectation step strictly retains the probabilistic assignment mechanism of the original framework. At each iteration, the current pathway weights are combined with the discretised path-specific probabilities to calculate the posterior responsibility of each latent transmission path (CP, PS, PT, and PQ) for every constructed ICC interval.

M-step: Parameter Updating and Zero-Day Handling. The maximization step preserves the original structural logic while introducing essential Log-Normal adaptations. First, the global pathway weights are updated by averaging their respective posterior responsibilities across all observations. Second, the algorithm retains the PS-only updating principle—restricting target SI parameter estimation to information weighted by PS responsibilities—but replaces the Normal-scale moment update with a log-scale update, which is then transformed back to the natural-scale mean and variance. To enable this logarithmic transformation without altering path classification, a decoupled zero-day handling strategy is

introduced: zero-day ICC intervals retain their exact discretised likelihood in the E-step but are assigned a zero-safe proxy strictly during the logarithmic M-step.

Convergence Assessment. Finally, the convergence criterion is adapted to jointly monitor changes in the target SI parameters and the pathway weights. This dual condition prevents the algorithm from terminating prematurely when the SI estimates appear locally stable but the underlying latent path allocations are still fluctuating.

3.3 Simulation study design

The primary aim of the simulation study is to evaluate whether the proposed Log-Normal extension can reliably recover the true mean and standard deviation of the underlying serial interval (SI) distribution. Unlike real outbreak data, where the true SI parameters and transmission pathways are unknown, simulation allows ICC intervals to be generated under a controlled data-generating mechanism based on the assumed four-pathway mixture framework. With known true parameters and transmission-pathway proportions, we can first assess the feasibility of parameter recovery under favourable, model-based conditions.

Beyond this initial assessment, we investigate how estimation performance varies across different epidemiological settings. In particular, infectious diseases may differ in the relative dispersion of their SI distributions, while households and small outbreaks may exhibit different transmission structures. We therefore vary both the SI dispersion level and the relative proportions of the four transmission pathways, with particular attention to the primary-secondary (PS) component, which provides the most direct information about the underlying SI distribution. This allows us to examine how differences in pathway overlap and the availability of direct SI information affect parameter recovery and algorithmic convergence.

Finally, we investigate the sensitivity of the estimation procedure to sample size. By varying the number of ICC intervals, we assess whether the method can maintain reliable parameter recovery when less information is available, particularly in small-sample settings.

Together, these simulation experiments evaluate the feasibility, accuracy, and numerical reliability of the proposed method under different SI distributions, transmission structures, and sample sizes.

3.3.1 Simulation scenarios

3.3.1.1 Disease characteristics: SI dispersion levels

To investigate how the characteristics of the underlying SI distribution affect estimation performance, three dispersion levels were selected, informed by reported SI characteristics of different infectious diseases[3]. Rather than reproducing exact empirical estimates, these settings represent plausible disease-related patterns and an additional extreme condition for stress testing.

The three settings are defined by the natural-scale SI mean m and standard deviation s. These symbols are used in the simulation and implementation sections in place of the equivalent model notation μ and σ, respectively:

$$(m, s) \in \{(15, 3), (2, 2), (5, 10)\}.$$

They represent increasing levels of relative dispersion, measured by the coefficient of variation c_v, defined as the SI standard deviation divided by its mean:

$$c_v = \frac{s}{m}.$$

- **Low dispersion ($m = 15$, $s = 3$):** This setting is inspired by chickenpox (varicella), for which previous studies have reported relatively long and narrowly distributed serial intervals[3]. With a coefficient of variation of 0.2, the SI distribution is relatively concentrated around its mean. Consequently, the pathway-specific ICC densities are expected to be more clearly separated, providing favourable conditions for identifying the underlying SI parameters.
- **Moderate dispersion ($m = 2$, $s = 2$):** This setting is motivated by influenza, which typically has a much shorter mean SI[3]. A relatively large standard deviation was selected to examine a more variable SI distribution, with a coefficient of variation of 1. The resulting greater right-skewness and overlap between pathway components are expected to make parameter recovery more challenging, particularly when distinguishing the primary-secondary component from other transmission pathways.
- **High dispersion ($m = 5$, $s = 10$):** This setting is not intended to represent a specific infectious disease but serves as an extreme stress-test scenario. With a coefficient of variation of 2, the underlying SI distribution is highly right-skewed with a pronounced right tail. This scenario is designed to assess whether substantial pathway overlap and extreme distributional variability create difficulties for parameter identification, approximation accuracy, and numerical convergence.

3.3.1.2 Transmission structure: pathway-weight configurations

In household and small-population outbreaks, the relative proportions of different transmission pathways may vary depending on the underlying transmission structure. To investigate how these variations affect SI parameter recovery, three pathway-weight profiles were designed to represent different outbreak structures. Particular attention was given to the primary-secondary (PS) pathway, as its ICC intervals directly correspond to the underlying SI distribution and therefore provide the most direct information for estimating the true SI mean and standard deviation.

Each SI dispersion setting was combined with the following three pathway-weight profiles. Here $w_{\text{CP}}, w_{\text{PS}}, w_{\text{PT}}$, and w_{PQ} are the mixture proportions assigned to the four named pathways and sum to one within each profile:

Table 3: Transmission-pathway weight profiles used in the simulation study.

Profile	w_{CP}	w_{PS}	w_{PT}	w_{PQ}
Baseline	0.20	0.50	0.20	0.10
CP-heavy	0.40	0.35	0.15	0.10
Higher-generation-heavy	0.10	0.35	0.35	0.20

The **baseline** profile represents the most favourable setting, with the highest proportion of PS observations ($w_{\mathrm{PS}} = 0.50$), which is expected to facilitate SI parameter recovery. In the **CP-heavy** profile, the PS proportion is reduced to 0.35, while the CP proportion increases to 0.40, representing outbreaks with a greater contribution from cases infected by a common source. In the **higher-generation-heavy** profile, the PS proportion is similarly reduced to 0.35, but the combined contribution of PT and PQ increases to 0.55, representing outbreaks with more transmission through intermediate cases.

By comparing these three profiles, we aim to assess whether the model can reliably recover the true SI mean and standard deviation when less direct PS information is available, and whether different compositions of the remaining transmission pathways introduce additional challenges to parameter identification.

However, the difficulty of distinguishing transmission pathways may also depend on the dispersion of the underlying SI distribution. To explore the combined effects of these two factors, we crossed the three pathway-weight profiles with the three SI dispersion levels, resulting in nine simulation scenarios. Figure 9 illustrates the corresponding theoretical ICC mixture densities, allowing us to visually assess the degree of overlap between pathway components and anticipate potential challenges in parameter recovery before examining the simulation results.

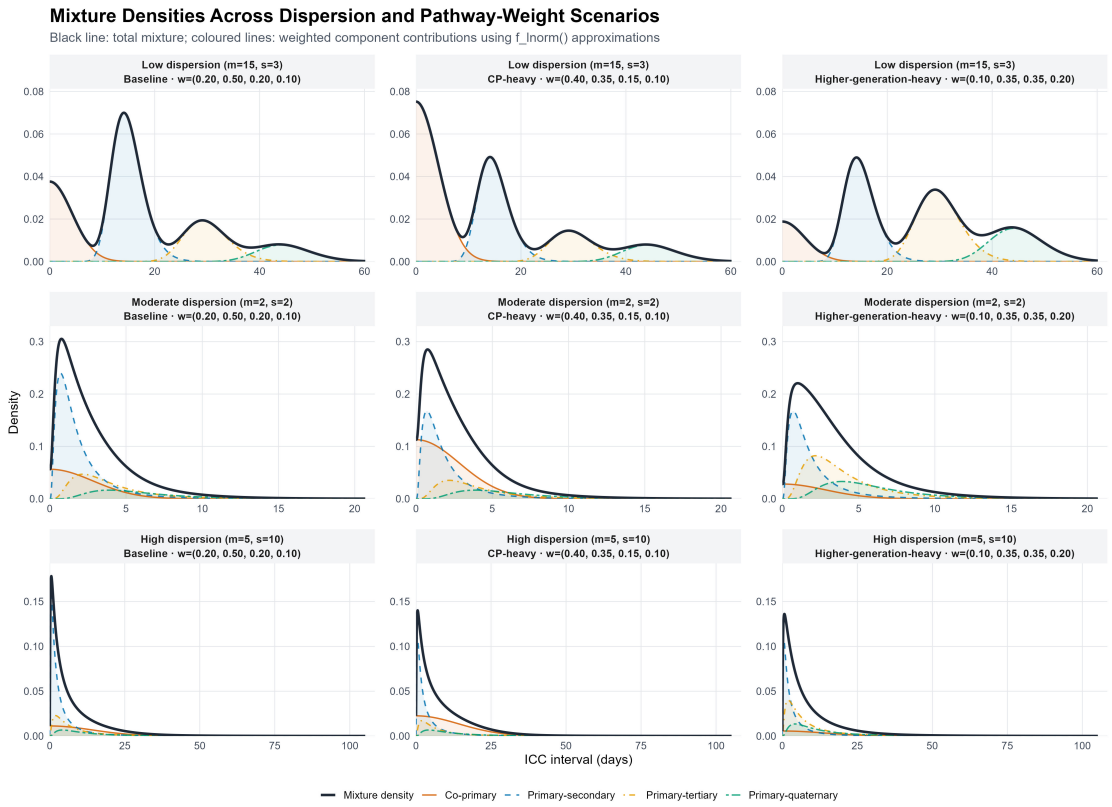


Figure 9: Theoretical ICC densities for the nine simulation scenarios. Rows vary the true SI mean and SD (low, moderate, high dispersion); columns vary the baseline, CP-heavy, and higher-generation-heavy pathway weights. Within each panel, the horizontal axis is ICC days and the vertical axis is density; the solid black curve is the total mixture and the coloured curves are weighted CP (orange), PS (blue), PT (yellow), and PQ (green) components. Horizontal scales differ by row. Compare component overlap and changes in mixture shape across scenarios; weights in each heading are ordered CP, PS, PT, PQ.

Under low dispersion (mean = 15, SD = 3), the pathway components are relatively well separated, particularly the primary-secondary component, suggesting more favourable conditions for recovering the underlying SI parameters. In contrast, under moderate dispersion (mean = 2, SD = 2), the component densities overlap substantially, making their individual contributions more difficult to distinguish. This challenge may become more pronounced under high dispersion (mean = 5, SD = 10), where the components are heavily concentrated at shorter ICC intervals and exhibit substantial overlap and extended right tails.

Across all dispersion levels, reducing the PS weight also decreases the relative contribution of the component that directly represents the underlying SI distribution. We therefore expect parameter recovery to become more challenging as component overlap increases or direct PS information decreases. These

expectations will be examined quantitatively through RMSE and convergence performance in the subsequent simulation results.

3.3.1.3 Information availability: sample size

To investigate how the amount of available ICC data affects estimation performance, four sample sizes were considered: $N \in \{20, 50, 100, 200\}$, where N denotes the total number of ICC intervals in one simulated dataset. This scalar use of N is distinct from the distribution notation $N(a, b)$, in which parentheses specify a Normal distribution. The earlier likelihood section uses the equivalent lower-case symbol n for a generic dataset size; upper-case N is retained here to match the simulation outputs and Appendix.

This analysis is particularly relevant to household and small-outbreak settings, where only a limited number of observations may be available. With smaller sample sizes, fewer observations contribute information about each latent transmission pathway, potentially making it more difficult for the model to distinguish the four mixture components and accurately estimate their relative contributions. In particular, limited information from the primary-secondary component may reduce the reliability of the estimated SI mean and standard deviation.

By comparing estimation performance across these sample sizes, we aim to assess the sensitivity of parameter recovery and algorithmic convergence to data sparsity, with particular attention to whether the method remains reliable under small-sample conditions.

3.3.2 Data generation and fitting protocol

For each simulation scenario, ICC interval data were generated according to the specified SI parameters and transmission-pathway weights. Primary-secondary (PS) intervals were sampled directly from the underlying Lognormal SI distribution, while primary-tertiary (PT) and primary-quaternary (PQ) intervals were generated as sums of two and three independent Lognormal variables, respectively. Co-primary (CP) intervals were generated from a half-Normal distribution. The number of observations assigned to each pathway was determined by rounding the product of the target sample size and the corresponding pathway weight. The generated continuous ICC intervals were then rounded to integer days to reflect the discrete nature of symptom-onset data.

The three SI dispersion settings and three transmission structures were combined with four sample sizes ($N = 20, 50, 100, 200$), resulting in 36 simulation configurations. Each configuration was evaluated using 100 independently generated datasets, with a fixed random seed of 2026 for reproducibility.

For model fitting, the empirical mean and standard deviation of each simulated ICC dataset were used as the initial values for the EM algorithm, rather than the true underlying SI parameters. This provides a more realistic initialization procedure, as the true parameters would be unknown in practical applications. The Lognormal mixture model was fitted with a maximum of 300 EM iterations, and the resulting parameter estimates and convergence information were recorded for subsequent evaluation.

3.3.3 Evaluation metrics

The root mean squared error (RMSE) was selected as the primary measure of parameter estimation accuracy, calculated separately for the SI mean and standard deviation. For each parameter $\theta \in \{m, s\}$, RMSE is defined as

$$\text{RMSE}(\hat{\theta}) = \sqrt{\frac{1}{R_{\text{conv},\theta}} \sum_{r \in \mathcal{C}_\theta} (\hat{\theta}_r - \theta)^2},$$

where θ denotes the true parameter value and is either the SI mean m or SD s; $\hat{\theta}_r$ is its estimate from simulation replicate r; $\mathcal{C}_\theta$ is the set of converged replicates with finite estimates for that parameter; and $R_{\text{conv},\theta} = |\mathcal{C}_\theta|$ is the number, or cardinality, of replicates in that set. Each scenario contains 100 repetitions; convergence percentages use all 100, whereas RMSE uses only the eligible converged estimates.

RMSE was chosen because it provides a single measure of overall estimation error, reflecting both systematic deviation from the true parameter and variability across simulation replicates. By assigning greater weight to larger errors, it is particularly useful for identifying unstable estimation performance under challenging scenarios. Furthermore, RMSE is expressed in the same units as the estimated SI parameters (days), allowing direct interpretation of estimation accuracy. Lower RMSE values indicate more accurate parameter recovery.

In addition to RMSE, algorithmic convergence was also assessed using the convergence rate and the number of EM iterations. This was intended to evaluate the computational efficiency of the estimation procedure, particularly whether it could achieve convergence within the maximum of 300 iterations and how quickly convergence was reached.

3.4 Data analysis

The simulation study evaluates parameter recovery because the true SI parameters and pathway composition are known by design. Real outbreak data provide a complementary question: whether the extended method can be applied successfully when the underlying transmission links and true SI distribution are unknown. The empirical analysis therefore assesses the practical behaviour of the Log-Normal implementation on observed, discrete ICC data and examines how its estimates and fitted mixture distribution compare with those from the existing Normal model.

To address this question, the Log-Normal extension was applied to four historical scabies datasets previously analysed under a Normal SI model: studies by Akunzirwe et al.[8], Ariza et al.[9], Kaburi et al.[10], and Tjon-Kon-Fat et al.[11]. These applications allow the natural-scale SI mean and standard deviation from the two distributional assumptions to be compared alongside their fitted ICC mixture curves. Similar estimates would provide evidence that the extension behaves consistently in an empirical setting, while

differences in the curves can show how the longer right tail of the Log-Normal assumption changes the representation of the ICC distribution.

Because the true transmission pairs and SI distributions in these outbreaks are not known, this comparison cannot establish parameter-recovery accuracy or determine that one distribution is universally preferable. Its purpose is instead to assess numerical stability, epidemiological plausibility, and sensitivity to the assumed SI distribution when the method is applied to real data.

Table 4: Estimated serial interval means and standard deviations (days) for four scabies outbreak datasets, as reported by[7] under the Normal SI model.

Study	Setting	Mean	Standard deviation
Akunzirwe et al.	Fishing community in Uganda	122.9	26.9
Ariza et al.	Kindergarten in Germany	98.4	8.5
Kaburi et al.	Pre-school in Ghana	167.3	9.7
Tjon-Kon-Fat et al.	Care home in the Netherlands	110.7	16.1

4 Results

4.1 Simulation

4.1.1 Main simulation analysis (N = 100)

The primary simulation analysis was conducted with a sample size of $N = 100$ and 100 repetitions per scenario. This reference setting was used to evaluate parameter recovery and convergence across the nine combinations of SI dispersion levels and transmission-pathway profiles. Estimation accuracy was assessed using the RMSE of the SI mean and standard deviation, alongside the convergence rate of the EM algorithm; the numerical results and bias distributions are reported in Table 5 and Figure 10, respectively.

The results are first examined to identify how SI dispersion and transmission structure affect estimation performance. The influence of sample size is then investigated separately by comparing the same nine scenarios across $N = 20, 50, 100,$ and 200. Detailed numerical results for all simulation configurations are provided in the Appendix.

Table 5: Parameter-recovery RMSE and EM convergence across nine simulation scenarios (N = 100; 100 repetitions per scenario). RMSE is calculated from finite converged estimates; convergence percentages use all repetitions.

SI dispersion (m, s)	Pathway profile	Mean RMSE (days)	SD RMSE (days)	Convergence (%)
Low (15, 3)	Baseline	0.66	0.43	100
Low (15, 3)	CP-heavy	11.73	0.99	100
Low (15, 3)	Higher-generation-heavy	14.49	10.96	100
Moderate (2, 2)	Baseline	0.50	0.93	37
Moderate (2, 2)	CP-heavy	0.60	0.84	32
Moderate (2, 2)	Higher-generation-heavy	1.05	0.84	24
High (5, 10)	Baseline	2.16	6.81	48
High (5, 10)	CP-heavy	2.27	7.27	53
High (5, 10)	Higher-generation-heavy	1.83	6.63	37

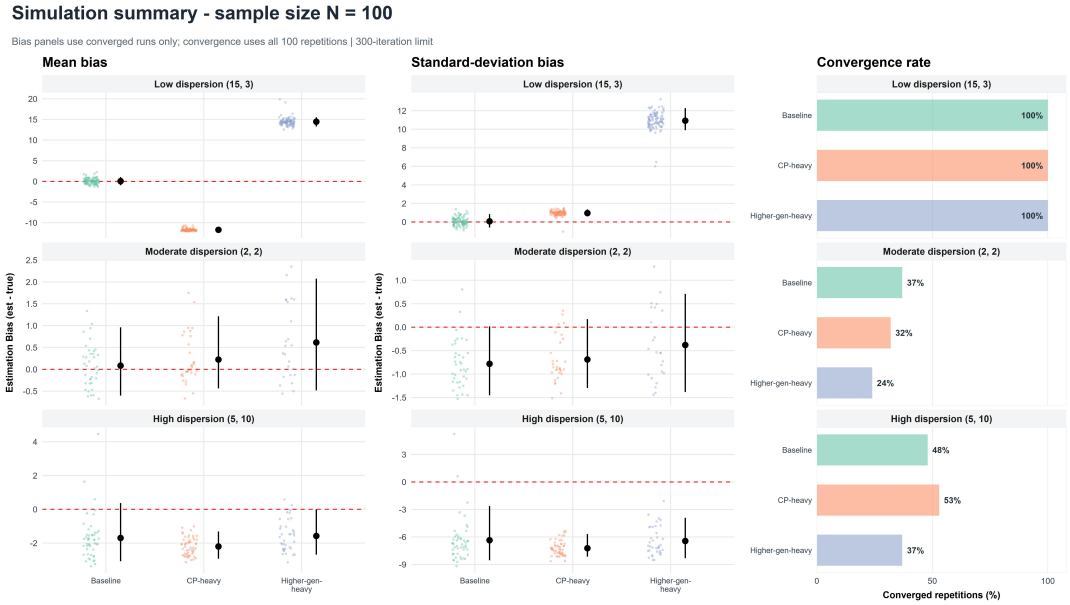


Figure 10: Simulation summary for N = 100 across three SI dispersion levels (rows) and pathway profiles (horizontal categories). The left and middle panels plot estimated minus true SI mean and SD in days for converged runs; coloured points are individual repetitions, black points are mean biases with 5th–95th percentile ranges, and red dashed lines mark zero bias. The right panels show the percentage of all 100 repetitions that converged, with profile on the vertical axis and percentage on the horizontal axis. The iteration cap is 300; the corresponding converged-run RMSE is reported in Table 5.

4.1.1.1 Effect of SI dispersion

As summarised in Table 5 and Figure 10, estimation performance at $N = 100$ varied substantially across the three SI dispersion levels. Under low dispersion ($m = 15$, $s = 3$), the baseline profile achieved mean and SD RMSE values of 0.66 and 0.43 days, respectively. These values indicate that the root-mean-squared deviations of the converged estimates from the true SI mean of 15 days and SD of 3 days were relatively small, suggesting accurate parameter recovery under this transmission structure. However, performance deteriorated considerably under the other pathway profiles. The mean RMSE increased to 11.73 days in the CP-heavy profile and 14.49 days in the higher-generation-heavy profile. Since the true SI mean was only 15 days, these errors were substantial relative to the target parameter, indicating poor mean recovery. The higher-generation-heavy profile also exhibited an SD RMSE of 10.96 days, considerably exceeding the true SD of 3 days. The bias distributions in Figure 10 suggest that these large errors were associated with systematic deviations from the true parameters rather than sampling variability alone.

Under moderate dispersion ($m = 2$, $s = 2$), Table 5 shows that the mean RMSE ranged from 0.50 to 1.05 days, while the SD RMSE ranged from 0.84 to 0.93 days. Although these absolute errors were smaller than those observed in several other scenarios, they were not negligible relative to the true SI mean and SD of 2 days. For example, a mean RMSE of 1.05 days represents a root-mean-squared estimation error exceeding half the true mean. Thus, the converged estimates showed moderate parameter-recovery accuracy, rather than uniformly small errors relative to the underlying SI parameters.

Under high dispersion ($m = 5$, $s = 10$), Table 5 shows that the mean RMSE ranged from 1.83 to 2.27 days, indicating substantial deviations from the true SI mean of 5 days. The SD RMSE was particularly large, ranging from 6.63 to 7.27 days against a true SD of 10 days. These results indicate considerable difficulty in recovering the variability of the underlying SI distribution. The poorer parameter recovery may be associated with the substantial overlap between pathway-specific distributions shown in Figure 10, which makes their contributions more difficult to distinguish.

Convergence performance also differed markedly across dispersion levels (Table 5; Figure 10). Under low dispersion, all three pathway profiles achieved 100% convergence within the 300-iteration limit, despite substantial estimation errors in two profiles. In contrast, convergence rates were lower under moderate dispersion (37%, 32%, and 24%) and high dispersion (48%, 53%, and 37%) for the baseline, CP-heavy, and higher-generation-heavy profiles, respectively. These results indicate that numerical convergence does not necessarily guarantee accurate parameter recovery. Conversely, the RMSE values for moderate and high dispersion describe only successfully converged runs, while many other runs did not meet the convergence criterion within 300 iterations. These scenarios may require more iterations to achieve convergence, although extending the iteration limit would not necessarily resolve the underlying estimation difficulties.

4.1.1.2 Effect of transmission structure

Transmission-pathway composition also influenced estimation accuracy, although its effect differed across dispersion levels (Table 5; Figure 10). Under low dispersion, the baseline profile, which contained the largest primary-secondary (PS) proportion ($w_{\mathrm{PS}} = 0.50$), achieved a mean RMSE of only 0.66 days. In comparison, reducing the PS proportion to 0.35 increased the mean RMSE to 11.73 days

in the CP-heavy profile and 14.49 days in the higher-generation-heavy profile. These substantially larger errors indicate that parameter recovery became much less accurate when direct PS information was reduced and other transmission pathways became more dominant.

Under moderate dispersion, Table 5 shows that the baseline profile again produced the lowest mean RMSE (0.50 days), compared with 0.60 and 1.05 days for the CP-heavy and higher-generation-heavy profiles, respectively. This suggests that a larger PS proportion may facilitate mean recovery. However, the corresponding SD RMSE values were 0.93, 0.84, and 0.84 days, respectively, indicating that the baseline profile did not provide the most accurate recovery of SI variability.

Under high dispersion, the relationship was less consistent (Table 5). The higher-generation-heavy profile produced the lowest mean RMSE (1.83 days), compared with 2.16 days for baseline and 2.27 days for CP-heavy. It also achieved the lowest SD RMSE (6.63 days), compared with 6.81 and 7.27 days for the other profiles. Nevertheless, these RMSE values remained substantial relative to the true SI mean of 5 days and SD of 10 days. Therefore, the comparatively lower RMSE under the higher-generation-heavy profile should not be interpreted as evidence of highly accurate parameter recovery.

Overall, the patterns in Table 5 and Figure 10 suggest that both the availability of direct PS information and the composition of the remaining transmission pathways affect estimation performance. A higher PS proportion appeared particularly beneficial for mean recovery under low and moderate dispersion, but this pattern did not hold consistently under high dispersion. Moreover, the substantial errors observed in the low-dispersion non-baseline profiles suggest that estimation difficulties cannot be explained by pathway overlap alone, motivating further investigation into the initialization of the EM algorithm.

4.1.2 Effect of sample size

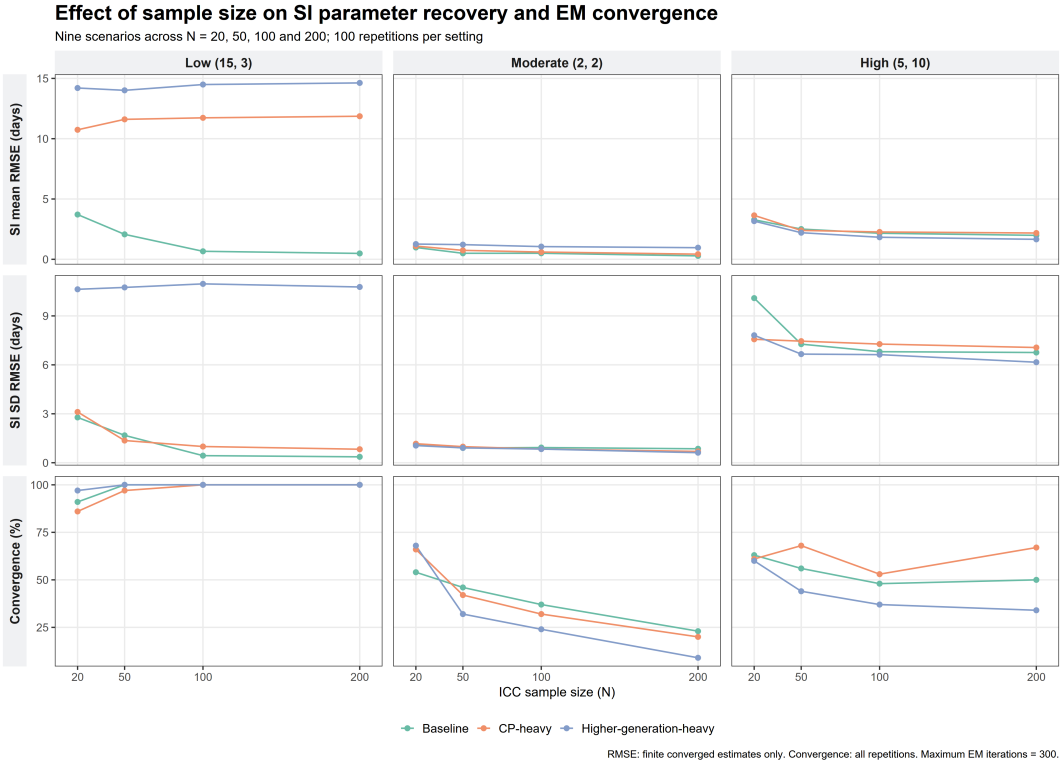


Figure 11: Effect of ICC sample size on parameter recovery and EM convergence. The horizontal axis gives sample size (20, 50, 100, 200); columns show low, moderate, and high SI dispersion, while rows show mean RMSE (days), SD RMSE (days), and convergence percentage. Colours identify the three pathway profiles. RMSE uses finite estimates from converged runs only; convergence uses all 100 repetitions per scenario. The EM iteration cap is 300. Compare how error and convergence change with sample size within each profile.

To investigate the influence of data availability, estimation performance was compared across four sample sizes ($N = 20, 50, 100$, and 200), while retaining the same nine combinations of SI dispersion and transmission structure. The resulting trends are shown in Figure 11, with the complete numerical results reported in Table 7.

As shown in Figure 11, increasing sample size tended to reduce RMSE, particularly under moderate and high dispersion, indicating that the SI parameter estimates from successfully converged runs generally became closer to their true values as more ICC observations were available. Under moderate dispersion, both mean and SD RMSE generally decreased as more ICC observations became available, although the reductions were relatively modest compared with those under high dispersion. Under high dispersion, estimation errors also decreased across all three pathway profiles, with the largest improvements

generally occurring between $N = 20$ and $N = 50$.

However, the trends in Figure 11 were not consistent across all scenarios. Under low dispersion, the baseline profile showed substantial reductions in both mean and SD RMSE as sample size increased. In contrast, the CP-heavy and higher-generation-heavy profiles continued to exhibit large mean RMSE values, with little improvement as N increased. In particular, the higher-generation-heavy profile maintained a mean RMSE of approximately 14–15 days across all four sample sizes, indicating a root-mean-squared estimation error comparable to the true SI mean of 15 days. Its SD RMSE also remained around 11 days, substantially exceeding the true SD of 3 days. These persistent errors suggest that increasing the sample size alone did not substantially improve parameter recovery in this scenario and may not resolve certain systematic estimation difficulties.

The relationship between sample size and convergence was more complex in Figure 11. Under low dispersion, convergence rates improved with increasing sample size and reached 100% for all three pathway profiles at $N = 100$ and $N = 200$. In contrast, under moderate dispersion, convergence rates consistently declined as sample size increased. For the baseline, CP-heavy, and higher-generation-heavy profiles, convergence decreased from 54%, 66%, and 68% at $N = 20$ to 23%, 20%, and 9% at $N = 200$, respectively. Under high dispersion, convergence patterns varied by transmission structure: the baseline and CP-heavy profiles fluctuated without a clear improvement, while the higher-generation-heavy profile showed a declining trend.

Taken together, Figure 11 and Table 7 indicate that larger sample sizes generally improve estimation accuracy in several scenarios, but do not necessarily lead to more reliable numerical convergence. In particular, the declining convergence rates under moderate dispersion, despite generally decreasing RMSE, suggest that estimation accuracy and convergence behaviour should be evaluated separately. Furthermore, the persistent estimation errors observed in some low-dispersion profiles indicate that certain limitations may not be overcome simply by increasing the amount of available ICC data.

4.1.3 Investigation of anomalous SI estimates under low-dispersion scenario

An unexpected finding from the main simulation was the substantial estimation error under low dispersion, despite consistently successful numerical convergence (Table 5; Figure 10). At $N = 100$, all three pathway profiles achieved 100% convergence, yet the SI mean RMSE reached 11.73 and 14.49 days under the CP-heavy and higher-generation-heavy profiles, respectively. The simulation summary plot also showed that the estimates were relatively tightly clustered around values substantially different from the true parameters. This suggests that the large errors were not simply due to high variability across repetitions, but reflected a tendency towards stable yet inaccurate estimates.

To investigate how these inaccurate estimates arose, Figure 12 presents representative EM fitting trajectories for the two non-baseline profiles, showing the evolution of the fitted mixture density, SI parameters, and pathway weights across iterations.

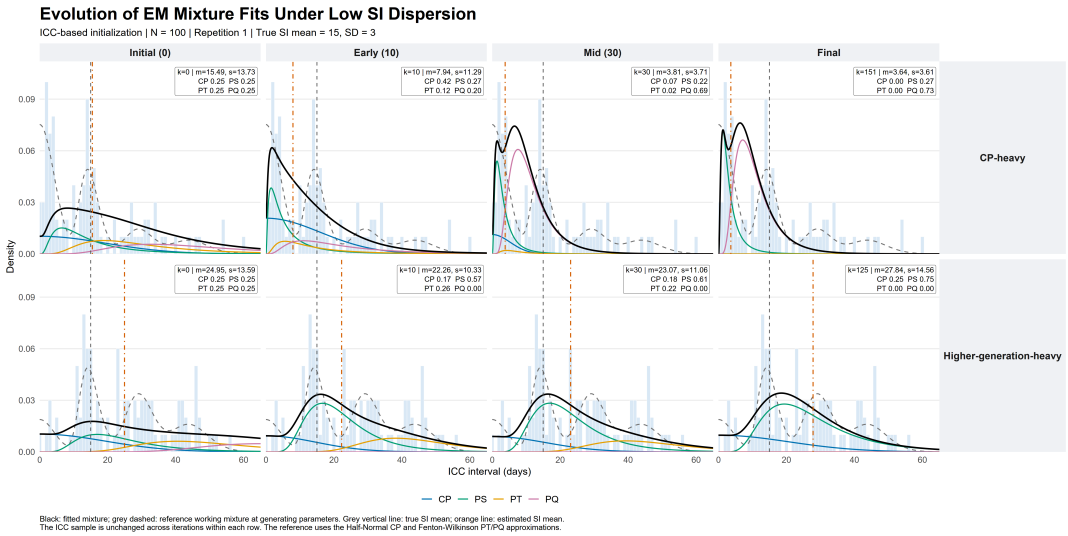


Figure 12: Representative EM trajectories under low SI dispersion ($m = 15$, $s = 3$) for one CP-heavy and one higher-generation-heavy repetition ($N = 100$), each initialized from empirical ICC moments. Columns show iterations 0, 10, 30, and the converged fit. The horizontal axis is ICC interval (days) and the vertical axis is density. Light-blue bars show the same ICC intervals throughout each row; coloured curves show weighted CP, PS, PT, and PQ densities, the black curve is their fitted mixture, and the grey dashed curve is the working mixture at the true parameters. Grey and orange vertical lines mark the true and estimated SI means. Compare how component weights and the estimated mean move across iterations.

In the CP-heavy trajectory in Figure 12, the estimated SI mean progressively decreased from 15.49 to 3.64 days, moving substantially away from the true value of 15 days. At the same time, the estimated CP weight approached zero, while the PQ weight increased to approximately 0.73. Notably, although the initial mean was close to the true SI mean, the initial SD was considerably larger than the true value (13.73 versus 3 days). This illustrates how an initially inaccurate representation of SI variability may be associated with changes in pathway allocation and subsequent parameter estimates.

In the higher-generation-heavy trajectory in Figure 12, a different pattern emerged. The estimated SI mean increased from 24.95 to 27.84 days, while the PS weight progressively increased to approximately 0.75 and the PT and PQ weights approached zero. In this case, both the initial mean and SD were substantially larger than their true values. As the algorithm proceeded, an increasing proportion of observations was assigned to the PS component, resulting in an estimated SI distribution that differed considerably from the underlying distribution used for data generation.

The trajectories in Figure 12 suggest that the EM procedure can converge to substantially different parameter estimates through changes in the allocation of observations among latent pathways. Because the estimation of SI parameters depends on the posterior responsibilities assigned to the PS component, inaccurate pathway allocation may reinforce subsequent parameter updates, allowing the algorithm to

stabilise at estimates far from the true values. Importantly, satisfying the numerical convergence criterion indicates that successive parameter updates have become sufficiently small; it does not establish that the recovered parameters are accurate.

The results therefore raise the possibility that the large estimation errors observed under low dispersion are partly associated with the initialization and numerical behaviour of the estimation procedure, rather than being an unavoidable consequence of the underlying SI distribution or transmission structure. This motivated the following sensitivity analysis, which investigates whether alternative initial parameter values can improve recovery of the true SI parameters.

4.2 Sensitivity analysis

4.2.1 Sensitivity to Initial Parameter Values

The main simulation study revealed an unexpected pattern under the low-dispersion setting, where the true serial interval (SI) mean and standard deviation were 15 and 3 days, respectively (Table 5; Figure 10). Although the proposed EM algorithm recovered the underlying SI parameters accurately under the baseline transmission-pathway profile, it produced substantially biased estimates under the CP-heavy and higher-generation-heavy profiles. In particular, the estimated SI means were consistently much lower or higher than the true value, even when the numerical convergence criterion was satisfied. This suggested that convergence of the iterative procedure did not necessarily imply accurate parameter recovery and motivated further investigation into the possible sources of these discrepancies.

One potential explanation was the choice of initial parameter values supplied to the EM algorithm. In the main simulation, the initial SI mean and standard deviation were taken directly from the empirical ICC intervals. However, because the ICC distribution is a mixture of four transmission pathways, its overall variability does not necessarily reflect that of the underlying SI distribution. Under low dispersion, the sample SD of the ICC intervals was substantially larger than the true SI SD in the representative trajectories shown in Figure 12, potentially affecting the initial component densities and subsequent posterior pathway responsibilities. Preliminary numerical comparisons in Figure 13 also suggested that changing the initial parameters, particularly the SD, could lead to substantially different final estimates.

To examine this possibility more systematically, a targeted sensitivity analysis was conducted under the low-dispersion setting. Three transmission-pathway profiles were retained: baseline, CP-heavy, and higher-generation-heavy. For each profile, 100 independent ICC datasets were generated with a sample size of $N = 100$. Each dataset was then fitted using four combinations of initial SI mean and SD:

1. **True-parameter start:** Both initial parameters were set to the true values used to generate the simulated SI distribution, $(m_0, s_0) = (m, s) = (15, 3)$. This represents an idealized initialization benchmark, rather than a strategy available in real outbreak applications.

2. **Mean-misspecified:** The initial mean was replaced by the empirical ICC sample mean, while the initial SD remained at its true value, $(m_0, s_0) = (\bar{T}, s)$. This isolates the effect of using an ICC-derived initial mean.

3. **SD-misspecified:** The initial mean was fixed at its true value, while the initial SD was replaced by the empirical ICC sample SD, $(m_0, s_0) = (m, s_T)$. This isolates the effect of using an ICC-derived initial SD.

4. **ICC-based:** Both parameters were initialized using the empirical mean and SD of the ICC intervals, $(m_0, s_0) = (\bar{T}, s_T)$. This reproduces the initialization approach adopted in the main simulation.

Here, m_0 and s_0 are the initial SI mean and SD supplied to the EM algorithm, while $\bar{T}$ and s_T denote the sample mean and SD of the simulated ICC observations. Importantly, the same ICC dataset was fitted under all four initialization strategies within each repetition, allowing direct comparisons without additional variation from generating different samples. The maximum number of EM iterations was set to 300, with the convergence tolerance unchanged.

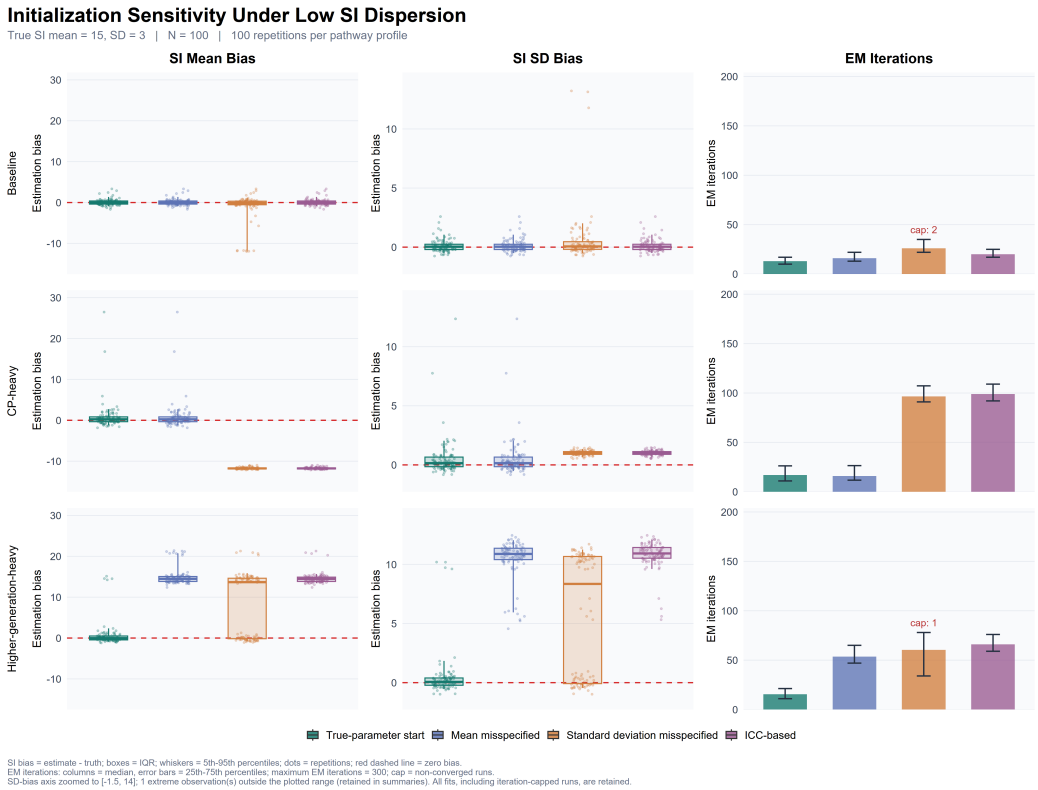


Figure 13: Sensitivity to four EM initialization strategies under low SI dispersion ($m = 15$, $s = 3$; $N = 100$; 100 repetitions per profile). Rows show baseline, CP-heavy, and higher-generation-heavy profiles; colours denote true-parameter, mean-misspecified, SD-misspecified, and ICC-based starts. The first two columns plot estimated minus true SI mean and SD (days): dots are repetitions, boxes show interquartile ranges, whiskers span the 5th–95th percentiles, and red dashed lines mark zero bias. The third column shows median EM iterations with interquartile-range bars; 'cap' counts runs reaching the 300-iteration limit. This diagnostic includes capped fits. The SD-bias axis is restricted for readability, excluding one extreme point from view while retaining it in summaries. Compare which initial parameter drives bias and computational effort.

As shown in Figure 13, **the influence of initial parameter values differed substantially across transmission-pathway profiles**. Under the baseline profile, parameter estimates were generally close to the true SI values across initialization strategies. The true-parameter, mean-misspecified, and ICC-based starts produced nearly identical mean estimates, averaging approximately 15.07 days. However, the SD-misspecified strategy occasionally produced markedly underestimated means, indicating that the baseline profile was not entirely insensitive to initialization.

The CP-heavy profile in Figure 13 exhibited a much stronger dependence on the initial SD. When the true SD was supplied, both the true-parameter and mean-misspecified starts produced estimates close to

the target distribution for most repetitions, with an average estimated SI mean of approximately 15.82 days. In contrast, the SD-misspecified and ICC-based strategies consistently produced substantial underestimation, with average estimated means of approximately 3.26 days, corresponding to a bias of -11.74 days. These inaccurate solutions were accompanied by pronounced deviations in the fitted pathway weights: the estimated CP weight approached zero, while the PQ weight increased to approximately 0.79, compared with its true value of 0.10. Notably, all runs under these two initialization strategies satisfied the numerical convergence criterion despite the substantial estimation bias.

A different pattern was observed for the higher-generation-heavy profile in Figure 13, where both the initial mean and SD affected parameter recovery. The true-parameter start produced an average estimated SI mean of 15.64 days. However, replacing the initial mean with the ICC sample mean resulted in an average estimate of approximately 29.93 days, even when the initial SD remained correctly specified. The SD-misspecified strategy produced an intermediate average of approximately 23.05 days, with final estimates concentrated in distinct regions near the true value and at substantially higher values. When both parameters were initialized using ICC sample moments, the estimated mean increased to approximately 29.62 days. These results suggest that the higher-generation-heavy profile was sensitive to both initial parameters, rather than predominantly to the initial SD.

Comparisons across pathway profiles in Figure 13 further illustrate that the effect of misspecification was not uniform. Mean misspecification had little effect on final estimates under the baseline and CP-heavy profiles but was associated with substantial overestimation under the higher-generation-heavy profile. Conversely, SD misspecification was particularly detrimental under the CP-heavy profile and produced less consistent outcomes under the other profiles. The ICC-based strategy performed well under the baseline profile but resulted in substantial biases under both alternative pathway compositions. Thus, the results indicate that the sensitivity of the estimation procedure to initialization is closely associated with the underlying pathway composition.

Initialization also influenced computational behaviour in Figure 13. Under the CP-heavy profile, median EM iterations increased from approximately 16–17 when the true SD was supplied to approximately 97–99 when using the ICC-derived SD. Similarly, under the higher-generation-heavy profile, the median iteration count increased from 15.5 with the true-parameter start to 66 with the ICC-based start. These differences suggest that less suitable starting values may lead to slower stabilization, even when the algorithm eventually satisfies its convergence criterion.

Overall, Figure 13 provides evidence that inaccurate parameter recovery under low dispersion can, at least partly, be associated with the initialization of the EM algorithm. In particular, changing the initial parameters substantially improved estimation accuracy under the CP-heavy and higher-generation-heavy profiles, demonstrating that the large biases observed in the main simulation were not inevitable under those pathway compositions. Nevertheless, the analysis was restricted to the low-dispersion setting and should be interpreted as a preliminary investigation of initialization sensitivity rather than a comprehensive assessment across all underlying SI distributions.

4.3 Data analysis

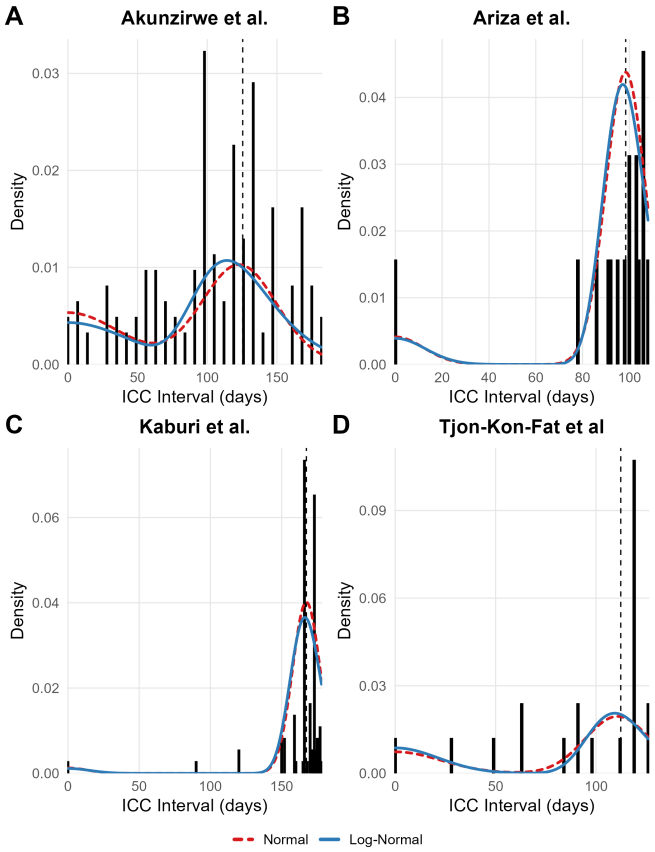


Figure 14: ICC distributions and fitted mixture densities for four scabies outbreak datasets: (A) Akunzirwe, (B) Ariza, (C) Kaburi, and (D) Tjon-Kon-Fat. The horizontal axis shows ICC interval (days) and the vertical axis shows density. Black bars represent the ICC distribution, while the red dashed and blue solid curves show the fitted Normal and Log-Normal mixture densities, respectively. The vertical black dashed line indicates the estimated SI mean under the Log-Normal model. The panels allow comparison of the two distributional assumptions against the ICC patterns.

Table 6: Comparison of estimated serial interval means and standard deviations (days) between the Normal and Lognormal models across four scabies outbreak datasets.

Study	Normal mean	Lognormal mean	Normal SD	Lognormal SD
Akunzirwe	122.9	125.4	26.9	31.0
Ariza	98.4	98.4	8.5	9.0
Kaburi	167.3	167.4	9.7	10.7

Study	Normal mean	Lognormal mean	Normal SD	Lognormal SD
Tjon-Kon-Fat	110.7	112.3	16.1	15.2

Figure 14 compares the fitted Normal and Lognormal ICC mixture distributions across the four scabies datasets, while Table 6 presents the corresponding SI parameter estimates. Overall, the two models produced broadly comparable estimates, with the Lognormal model yielding slightly higher SI means in three datasets and an identical mean in the remaining dataset. This tendency may reflect the right-skewed shape of the Lognormal distribution, which allows the underlying PS component to accommodate a longer right tail. Consequently, some longer ICC intervals may be explained differently under the two distributional assumptions, potentially contributing to the slightly higher SI mean estimates. However, this interpretation remains tentative, as the fitted mixture curves do not directly reveal the posterior pathway assignments.

For the Akunzirwe et al. dataset (Panel A of Figure 14), the ICC intervals were relatively widely dispersed, and the two fitted curves showed more noticeable differences than in the other datasets. As reported in Table 6, the estimated SI mean increased from 122.9 days under the Normal assumption to 125.4 days under the Lognormal assumption, while the SD increased more substantially from 26.9 to 31.0 days. The Lognormal mixture exhibited a slightly broader main peak and a lower fitted density than the Normal mixture at shorter ICC intervals, where the CP component is expected to contribute. These differences suggest that the two models may represent the shorter intervals and the main concentration of observations differently. The larger Lognormal SD may partly reflect this difference in the fitted mixture structure, although the relative contributions of individual pathways cannot be determined from the overall curves alone.

For the Ariza et al. dataset (Panel B of Figure 14), the ICC intervals were strongly concentrated around 90–105 days, and the fitted Normal and Lognormal curves closely overlapped around the main peak. Both models estimated an identical SI mean of 98.4 days, with only a small increase in SD from 8.5 to 9.0 days under the Lognormal assumption (Table 6). Similarly, in the Kaburi et al. dataset (Panel C of Figure 14), both models captured the pronounced concentration of ICC intervals around 160–175 days. The estimated SI means were nearly identical (167.3 and 167.4 days), while the SD increased slightly from 9.7 to 10.7 days (Table 6). These results indicate that changing the SI distributional assumption had relatively little influence on parameter estimates in these two datasets.

For the Tjon-Kon-Fat et al. dataset (Panel D of Figure 14), the fitted Normal and Lognormal mixture curves were also very similar across the ICC range. As reported in Table 6, the estimated SI mean increased slightly from 110.7 to 112.3 days under the Lognormal assumption, whereas the estimated SD decreased from 16.1 to 15.2 days. Unlike the other three datasets, the Lognormal model therefore produced a marginally smaller estimate of SI variability, although the overall differences remained modest.

Taken together, Figure 14 and Table 6 demonstrate that the Lognormal extension can produce SI parameter estimates broadly consistent with those obtained under the established Normal model. Despite some differences in the fitted mixture shapes, particularly in the Akunzirwe dataset, the two models

generally provided similar representations of the ICC timing patterns. This agreement provides additional support for the practical feasibility of the Lognormal extension and increases confidence in its application to real outbreak data. Nevertheless, because the true SI distributions are unknown, these findings do not establish that the Lognormal assumption provides more accurate estimates or a superior fit. Rather, the extension offers an additional distributional option for analysing diseases whose underlying serial intervals may be positive and right-skewed.

5 Software Implementation

5.1 Package context and purpose

The proposed Lognormal extension was implemented within the existing `{mitey}` R package, originally developed to support serial-interval estimation and reproduction-number analysis using infectious-disease outbreak data. The package has two main functions: estimating the mean and standard deviation of the serial interval (SI) from index-case-to-case (ICC) intervals using the transmission-pathway framework of Vink et al., and estimating the time-varying reproduction number using the Wallinga–Lipsitch method. This project focuses exclusively on extending the SI estimation component, leaving the reproduction-number functionality unchanged.

The original `si_estim()` function supports Normal and Gamma assumptions for the underlying SI distribution. This project introduces a Lognormal option to accommodate right-skewed SI distributions. Rather than replacing the existing models, the extension provides greater flexibility in selecting an appropriate distributional assumption while preserving the original workflow of fitting ICC intervals and estimating SI parameters and transmission-pathway proportions.

5.2 Input layer: adding the Lognormal option

The primary user-facing modification is the addition of `dist = "lognormal"` to `si_estim()`, alongside the existing `"normal"` and `"gamma"` options. The input remains a numeric vector of ICC intervals measured in days. Negative intervals are not permitted under the Lognormal specification, while zero-day observations are retained to account for cases with symptom onset recorded on the same day.

The existing estimation controls, including the maximum number of EM iterations, initial SI mean and standard deviation, convergence tolerance, and multiple starting values, remain available. Thus, all three distributional assumptions can be fitted through the same function interface.

5.3 Mixture-model layer: pathway-specific densities

Introducing the Lognormal assumption required modifications to the four pathway-specific density functions. The primary-secondary (PS) component follows the underlying Lognormal SI distribution

exactly, while the co-primary (CP) component is represented by a Half-Normal working approximation. The primary-tertiary (PT) and primary-quaternary (PQ) components involve sums of two and three independent Lognormal variables, respectively. Since these sums lack convenient closed-form densities, Fenton–Wilkinson Lognormal approximations were introduced to match their natural-scale means and variances.

These component densities were incorporated into the mixture likelihood and the functions used to evaluate discrete ICC observations. Because symptom-onset dates are recorded in calendar days, the component densities are numerically integrated over the corresponding daily observation intervals. This preserves the original discretisation framework while accommodating the new distributional assumption.

5.4 Estimation layer: adapting the EM algorithm

The expectation–maximization (EM) procedure was adapted to improve pathway assignment, numerical stability, and convergence assessment.

First, the E-step was modified to calculate posterior responsibilities using the current mixture weights. A responsibility represents the estimated probability that an individual ICC observation belongs to a particular latent transmission pathway. Unlike the previous implementation, which normalised pathway-specific probabilities without accounting for their relative prevalence, the updated calculation uses the weighted sum of probabilities across all four pathways in the denominator. This ensures that the responsibilities reflect both the compatibility of each observation with a pathway and its estimated prevalence in the mixture. The resulting responsibilities are averaged to update the pathway weights, while the underlying Lognormal SI parameters are updated using PS-weighted log-scale moments, subsequently transformed back to the natural scale.

Second, an E/M-decoupled strategy was introduced to accommodate zero-day ICC observations, which occur when two cases have symptom onset recorded on the same day. These observations retain their original zero-day values when calculating discretised probabilities and posterior responsibilities in the E-step. However, because the M-step requires logarithmic transformations, zero values are replaced by a value of one only during the log-scale parameter update. This avoids the undefined $\log(0)$ and prevents extreme negative log values from distorting the estimated SI parameters, particularly the variance, while preserving the original observations for pathway assignment.

Finally, the convergence criterion was strengthened to account for changes in both SI parameters and pathway weights. Previously, convergence was assessed based on changes in the estimated SI mean and standard deviation between successive iterations. The updated procedure additionally requires the maximum absolute change across the four pathway weights to fall below a specified tolerance. Convergence is therefore declared only when the relative changes in the SI mean and standard deviation, together with the absolute changes in pathway weights, satisfy the convergence criterion. This reduces the risk of premature termination when SI parameter estimates appear stable but the underlying pathway allocations are still changing.

5.5 Output layer

The extended implementation preserves the existing output structure. The fitted object returns the estimated natural-scale SI mean and standard deviation, the four pathway weights, convergence status, number of iterations, fitted log-likelihood, and iteration history.

The plotting functions were also extended to support the Lognormal option, allowing the fitted mixture density and its pathway-specific components to be visualised against the ICC distribution. These modifications maintain consistency with the original package while supporting additional distributional assumptions.

5.6 Use in simulation and empirical analysis

The extended implementation was evaluated through simulation studies combining three SI dispersion levels, three transmission-pathway configurations, and four sample sizes. The simulation results were used to assess parameter recovery using RMSE, together with algorithmic convergence and sensitivity to initialization.

The method was subsequently applied to four historical scabies datasets previously analysed using the Normal SI model. The empirical analysis compared estimated SI parameters and fitted mixture densities under the Normal and Lognormal assumptions. As the true SI distributions are unknown, this comparison provides a practical assessment of numerical performance and epidemiological plausibility rather than direct evidence of parameter-recovery accuracy.

6 Discussion

This study extended the transmission-pathway mixture framework of Vink et al.[3] by introducing a Lognormal assumption for the underlying serial interval (SI) distribution. While the original framework and subsequent implementations primarily considered Normal and Gamma distributions, the proposed extension provides an additional option for infectious diseases with long-tailed and right-skewed serial intervals. The extension was implemented within the `{mitey}` R package and evaluated through simulation studies covering different SI dispersion levels, household and small-outbreak transmission structures, and sample sizes, followed by applications to four historical scabies datasets. Overall, the findings demonstrate the feasibility of the Lognormal extension while highlighting the epidemiological conditions under which reliable SI parameter recovery may be achieved.

The simulation results in Table 5 and Figure 10 suggest that the proposed method is most promising for infectious diseases with relatively low or moderate SI dispersion, particularly in household or small-population outbreaks where primary-secondary (PS) transmission contributes a substantial proportion of the ICC intervals. Under the low-dispersion baseline scenario, where PS observations accounted for 50% of the mixture, the method achieved RMSE values of 0.66 and 0.43 days for the SI mean and standard deviation, respectively, with 100% convergence. However, estimation accuracy

deteriorated considerably when the PS proportion decreased and other pathways became more dominant, even under the same low-dispersion setting. Under moderate dispersion, the converged estimates generally showed relatively low RMSE, although many runs did not satisfy the convergence criterion within 300 iterations, suggesting that more iterations may be required in some cases. In contrast, the high-dispersion scenarios exhibited larger estimation errors and substantial overlap between pathway components. These findings indicate that the applicability of the method depends not only on the SI characteristics of the infectious disease but also on whether the transmission structure of the observed outbreak provides sufficient direct PS information.

Another finding is the sensitivity of parameter recovery to the choice of initial values, as shown in Figure 12 and Figure 13. The low-dispersion simulations revealed that the algorithm could satisfy its numerical convergence criterion while producing estimates substantially different from the true SI parameters. Further investigation showed that initializing the SI parameters using the empirical mean and standard deviation of the entire ICC dataset could be problematic, particularly when the variability of the ICC mixture substantially exceeded that of the underlying SI distribution. The subsequent sensitivity analysis demonstrated that alternative starting values could substantially improve parameter recovery in some of these scenarios, with the effects depending on the transmission-pathway composition. This highlights the importance of selecting epidemiologically plausible initial parameters rather than relying exclusively on ICC sample moments. Previous applications of the Vink framework have also recognised the importance of initialization. In practical applications, researchers may draw on previous studies or disease-specific epidemiological knowledge to select reasonable starting values, potentially considering multiple initializations to assess the stability of the resulting estimates.

Finally, the application to four historical scabies datasets, summarised in Figure 14 and Table 6, provided an empirical assessment of the Lognormal extension. Across the datasets, the estimated SI means and standard deviations were broadly comparable to those obtained under the original Normal assumption, although some differences were observed in the fitted mixture densities. These findings suggest that the Lognormal extension can produce epidemiologically plausible estimates when applied to real outbreak data, supporting its practical applicability beyond simulated settings. However, because the true SI distributions and transmission links are unknown, this agreement does not establish that the Lognormal assumption provides more accurate estimates or a better fit than the Normal model. Rather, the extension offers an additional distributional option whose suitability should be assessed in relation to the characteristics of the disease and the available outbreak data.

6.1 Limitations

Despite demonstrating the feasibility of the Lognormal extension, the proposed method has several limitations that may affect the accuracy of SI parameter estimation and its applicability to different infectious diseases and outbreak settings.

6.1.1 Restricted transmission-pathway structure

The proposed method retains the four transmission pathways considered in the original Vink framework: co-primary (CP), primary-secondary (PS), primary-tertiary (PT), and primary-quaternary (PQ). While this structure provides a reasonable representation of transmission in households or small outbreak clusters, real-world outbreaks may involve more complex transmission networks, including additional transmission generations and multiple external introductions.

When the true transmission structure extends beyond these four pathways, some ICC intervals may not be adequately represented by the assumed mixture model. Consequently, the model may incorrectly attribute these observations to the existing components, potentially affecting the estimated pathway proportions and underlying SI parameters. The applicability of the method therefore depends on whether the four-pathway framework adequately represents the transmission structure of the outbreak being studied.

6.1.2 Pathway overlap, initialization, and practical identifiability

A key limitation of the proposed method is its reliance on the posterior responsibilities assigned to the latent transmission pathways during the E-step. Since the underlying SI parameters are updated using information weighted by the primary-secondary (PS) responsibilities, difficulties in distinguishing transmission pathways may affect subsequent parameter updates and lead to inaccurate SI estimates. However, such difficulties can arise for different reasons, with important implications for whether they can be overcome.

The first concerns the inherent overlap between pathway-specific distributions. For infectious diseases with relatively high SI dispersion, the four transmission-pathway components may overlap substantially even when the true SI parameters are used. Under these conditions, the ICC intervals may provide insufficient information to distinguish the contributions of different pathways, weakening the practical identifiability of the underlying SI parameters. This represents a fundamental challenge to the applicability of the method for highly dispersed SI distributions, which cannot necessarily be resolved by selecting better initial values or allowing more EM iterations.

The second concerns sensitivity to initialization. Even when the true pathway distributions are relatively well separated, unsuitable initial SI parameters may produce fitted component distributions with substantial overlap, leading to inaccurate pathway responsibilities during the E-step. These responsibilities may then influence subsequent parameter updates, causing the algorithm to converge to stable but inaccurate estimates. Unlike the inherent overlap associated with high SI dispersion, this difficulty may be mitigated through more appropriate initialization. As demonstrated by the sensitivity analysis, alternative starting values substantially improved parameter recovery in some of the problematic low-dispersion scenarios.

Therefore, it is important to distinguish between difficulties arising from limited information in the underlying ICC mixture and those introduced by unsuitable initial parameter values. While improved

initialization may enhance estimation performance in otherwise favourable settings, it cannot guarantee reliable parameter recovery when the transmission pathways are inherently difficult to distinguish.

6.1.3 Distributional and observational assumptions

The proposed extension assumes that the underlying SI follows a Lognormal distribution, which is strictly positive and right-skewed. Unlike the Normal distribution, which is symmetric and allows negative SI values, the Lognormal model is therefore intended for infectious diseases whose serial intervals are expected to be positive and exhibit a relatively long right tail. This assumption may be inappropriate for diseases where negative serial intervals occur or where the underlying SI distribution has a substantially different shape.

The framework also assumes that individual serial intervals are independent and identically distributed (i.i.d.) and that the ICC intervals are statistically independent. In reality, SI characteristics may vary across individuals due to factors such as age, individual susceptibility, disease progression, or public health interventions. ICC observations from the same household may also be correlated because they share a common index case and transmission environment. However, such individual-level and transmission-related information is not always available in routine surveillance data. The i.i.d. assumption therefore provides a practical simplification, although unaccounted heterogeneity and dependence may affect parameter estimation.

Finally, the discretisation model assumes that the exact symptom-onset time is uniformly distributed within each reported calendar day. In practice, reported onset dates may also be affected by recall errors, inaccurate recording, or other measurement errors. These departures from the assumed observation process may introduce additional uncertainty or bias into the estimated SI distribution.

6.2 Future work

Several directions could further improve the flexibility and applicability of the proposed framework.

First, future research could extend the underlying SI distributional assumptions beyond the existing Normal, Gamma, and Lognormal options. As the shape and variability of serial intervals may differ across infectious diseases, no single distribution is expected to adequately represent all epidemiological settings. Exploring additional distribution families, such as Weibull or generalised Gamma distributions, could broaden the applicability of the framework to diseases with different SI characteristics.

Second, future work should develop systematic criteria to help researchers select an appropriate SI distribution for a given outbreak dataset. Although the `{mitey}` package now supports multiple distributional assumptions, determining which model best represents the underlying SI remains challenging because the true transmission links and SI distribution are generally unknown. Potential approaches include evaluating goodness-of-fit methods, such as smooth tests, alongside methods for comparing candidate models based on their fitted ICC mixture distributions. Such criteria could provide more

objective guidance for distribution selection, rather than relying solely on prior assumptions or visual comparisons of fitted curves.

Finally, the transmission-pathway framework could be extended to accommodate more complex outbreak structures, including transmission chains beyond the current four-pathway specification. An important challenge is that long ICC intervals may result either from a long-tailed underlying SI distribution or from transmission through additional generations. If the model does not adequately represent the true transmission structure, it may incorrectly attribute longer ICC intervals to the SI distribution, potentially biasing parameter estimates. Future research should therefore investigate how to determine when additional transmission pathways are epidemiologically warranted and how to incorporate them while maintaining model identifiability. Information about household or outbreak-cluster structure may provide useful guidance for assessing the plausibility of longer transmission chains.

7 Appendix

7.1 Appendix A. Additional Details of the EM Estimation Procedure

This appendix provides the mathematical updates and numerical details underlying the EM procedure described in the Methods section. The four transmission pathways, their Log-Normal working densities, and the triangular discretisation of ICC intervals are defined in the main text and are not repeated here. The additional details concern latent pathway responsibilities, PS-weighted parameter updates, zero-day observations, and convergence assessment.

7.1.1 A.1 Latent Pathway Responsibilities and the Vink Updating Principle

Throughout this Appendix, N is the number of ICC observations, $i \in \{1, \dots, N\}$ indexes observations, g and h index pathways in $\mathcal{G}$, and $r = 0, 1, 2, \dots$ indexes EM iterations. The vector $\mathbf{w} = (w_g : g \in \mathcal{G})$ contains the four pathway weights.

Let $T_i = t_i$ denote the observed integer-day ICC interval for observation i, and let $Z_i \in \mathcal{G} = \{\mathrm{CP}, \mathrm{PS}, \mathrm{PT}, \mathrm{PQ}\}$ denote its unobserved transmission pathway. Define the latent indicator z_{ig}, which records whether observation i belongs to pathway g:

$$z_{ig} = \begin{cases} 1, & Z_i = g, \\ 0, & Z_i \neq g. \end{cases}$$

If the pathway labels were observed, the complete-data log-likelihood associated with pathway membership would be

$$\ell_c(m, s, \mathbf{w}) = \sum_{i=1}^{N} \sum_{g \in \mathcal{G}} z_{ig} \left\{ \log w_g + \log p_{ig}(m, s) \right\},$$

where ℓ_c denotes the complete-data log-likelihood, the subscript c denotes "complete data", m and s denote the natural-scale SI mean and SD, w_g is the mixture weight of pathway g, and $p_{ig}(m, s)$ is the discretised probability of observation i under pathway g. The logarithm is the natural logarithm, and the two summation signs combine contributions over all observations and pathways.

At iteration r, the E-step replaces the unobserved indicators with the posterior responsibilities $\tau_{ig}^{(r)}$, the conditional probability that pathway g generated observation i under the current parameter values:

$$\tau_{ig}^{(r)} = \Pr(Z_i = g \mid T_i = t_i, \mathbf{w}^{(r)}, m^{(r)}, s^{(r)}) = \frac{w_g^{(r)} p_{ig}(m^{(r)}, s^{(r)})}{\sum_{h \in \mathcal{G}} w_h^{(r)} p_{ih}(m^{(r)}, s^{(r)})}.$$

In the denominator, h is a dummy pathway index used to sum over all four possible pathways. The responsibilities satisfy $\sum_g \tau_{ig}^{(r)} = 1$ for each observation and provide fractional rather than deterministic pathway assignments. Their average updates the pathway proportions:

$$w_g^{(r+1)} = \frac{1}{N} \sum_{i=1}^{N} \tau_{ig}^{(r)}.$$

The current weights must enter each E-step because pathway prevalence influences the posterior assignments. Importantly, all four pathways contribute to this weight update, whereas the parameters of the underlying SI distribution are updated using the primary-secondary (PS) responsibilities only. In the original Normal version of the Vink procedure[3], $\mu^{(r+1)}$ and $(\sigma^2)^{(r+1)}$ denote the updated natural-scale SI mean and variance. Their corresponding PS-weighted updates were

$$\mu^{(r+1)} = \frac{\sum_i \tau_{i,\mathrm{PS}}^{(r)} t_i}{\sum_i \tau_{i,\mathrm{PS}}^{(r)}}, \qquad (\sigma^2)^{(r+1)} = \frac{\sum_i \tau_{i,\mathrm{PS}}^{(r)} (t_i - \mu^{(r+1)})^2}{\sum_i \tau_{i,\mathrm{PS}}^{(r)}}.$$

In these expressions, each unbounded sum over i runs over all N observations. The expressions are included for comparison: the Log-Normal extension retains the PS-only updating principle but changes the scale on which the SI moments are calculated.

7.1.2 A.2 PS-Weighted Updates Under the Log-Normal SI Model

For the Log-Normal extension, write $\log X \sim N(\eta, \zeta^2)$ for a direct SI X, where η and ζ^2 are the mean and variance of the log-transformed SI. Let $a_i^{(r)} = \tau_{i,\mathrm{PS}}^{(r)}$ be observation i's PS responsibility, and define $Y_i = \log T_i^*$ as its zero-safe log interval, where T_i^* is the zero-safe M-step value described below. Set

$$A_1^{(r)} = \sum_{i=1}^{N} a_i^{(r)}, \qquad A_2^{(r)} = \sum_{i=1}^{N} \left(a_i^{(r)}\right)^2.$$

Thus, $A_1^{(r)}$ is the total PS responsibility at iteration r, while $A_2^{(r)}$ is the sum of squared PS responsibilities used in the finite-sample correction.

The responsibility-weighted log-scale mean and variance are updated by

$$\eta^{(r+1)} = \frac{\sum_{i=1}^{N} a_i^{(r)} Y_i}{A_1^{(r)}},$$

$$(\zeta^2)^{(r+1)} = \frac{A_1^{(r)}}{\left(A_1^{(r)}\right)^2 - A_2^{(r)}} \sum_{i=1}^{N} a_i^{(r)} \left(Y_i - \eta^{(r+1)}\right)^2.$$

The second expression is the finite-sample-corrected weighted variance used in the implementation; it is distinct from the uncorrected weighted variance commonly used in a likelihood M-step. The estimates are transformed back to the natural scale as follows, where $\exp(\cdot)$ denotes the exponential function:

$$m^{(r+1)} = \exp\left(\eta^{(r+1)} + \frac{(\zeta^2)^{(r+1)}}{2}\right),$$

$$s^{(r+1)} = \sqrt{\left\{\exp\left((\zeta^2)^{(r+1)}\right) - 1\right\} \exp\left\{2\eta^{(r+1)} + (\zeta^2)^{(r+1)}\right\}}.$$

The PS component represents a single direct SI and therefore follows the target Log-Normal distribution without an additional pathway-density approximation. Restricting the parameter updates to observations weighted by their PS responsibilities limits the direct influence of CP observations and longer-generation PT/PQ intervals. These other pathways still affect the estimates indirectly through their contributions to the E-step denominator and the subsequent PS responsibilities. In particular, misallocation of higher-generation observations to PS can influence the SI estimates despite the PS-only update.

Because this implementation uses corrected responsibility-weighted moments of discretised ICC values rather than an exact maximisation of the full interval-censored mixture likelihood, it should be understood as a Vink-style EM updating procedure rather than a guarantee of monotonic improvement in the full observed-data likelihood.

7.1.3 A.3 Treatment of Zero-Day ICC Intervals

Zero-day ICC intervals represent valid observations in which cases have symptom onset recorded on the same day. However, extending the Vink updating procedure to a Log-Normal SI distribution introduces a numerical difficulty because the M-step operates on logarithmically transformed intervals, for which

$$\log(0) = -\infty.$$

A naive approach would be to replace zero with a very small positive value. However, this produces an extremely negative log observation. If the corresponding interval receives non-negligible PS responsibility, it may pull the estimated log-scale mean downward and substantially inflate the log-scale variance. The resulting PS density becomes more dispersed, potentially altering the posterior responsibilities calculated in subsequent E-steps. These changes may, in turn, further distort the parameter updates, creating a positive-feedback mechanism:

$$\begin{aligned} T_i = 0 &\longrightarrow \text{extreme negative log value} \\ &\longrightarrow \text{inflated log-scale variance} \\ &\longrightarrow \text{overly dispersed PS density} \\ &\longrightarrow \text{distorted PS responsibilities} \\ &\longrightarrow \text{potential further variance inflation.} \end{aligned}$$

To avoid this numerical instability, an **E/M-decoupled zero-handling strategy** was implemented. In the E-step, the original zero-day observation is retained when evaluating the pathway-specific discretised likelihood. In the current implementation, its contribution is calculated as

$$p_g^{\text{impl}}(0; m, s) = \int_0^1 2(1-c) f_g(c; m, s)\, dc.$$

Here, $p_g^{\text{impl}}(0; m, s)$ is the implemented probability assigned to a zero-day observation under pathway g, the superscript "impl" distinguishes this implementation-specific expression from the general discretisation formula, $f_g(c; m, s)$ is pathway g's continuous ICC density at c, and m, s are the natural-scale SI mean and SD.

Only during the logarithmic M-step is a separate zero-safe working value introduced:

$$T_i^* = \begin{cases} 1, & T_i = 0, \\ T_i, & T_i > 0, \end{cases} \qquad Y_i = \log(T_i^*).$$

Thus, a zero-day observation contributes $Y_i = 0$ to the responsibility-weighted log-scale parameter updates, avoiding the extreme negative log values that could otherwise destabilise estimation. The replacement value of one is a computational proxy rather than an assumption that the underlying continuous ICC interval lasted exactly one day.

This separation retains the observed zero-day interval during pathway classification while preventing the logarithmic singularity from entering the M-step. Although the proxy does not constitute an exact treatment of interval censoring in the parameter update, it provides a numerically stable implementation of the PS-weighted updating procedure.

7.1.4 A.4 Initialization and Joint Convergence Criterion

The implementation begins with initial natural-scale SI parameters $(m^{(0)}, s^{(0)})$ and equal initial pathway weights $w_g^{(0)} = 1/4$. Unless alternative starting parameters are supplied, the initial mean and SD are obtained from the empirical ICC intervals. The sensitivity analysis in the Results section examines the effect of varying these initial values while keeping the ICC datasets fixed.

At iteration r, the stopping rule monitors relative changes in the natural-scale SI parameters and the largest absolute change in the four mixture weights:

$$\Delta_m^{(r)} = \frac{|m^{(r+1)} - m^{(r)}|}{|m^{(r)}| + \epsilon}, \qquad \Delta_s^{(r)} = \frac{|s^{(r+1)} - s^{(r)}|}{|s^{(r)}| + \epsilon},$$

$$\Delta_w^{(r)} = \max_{g \in \mathcal{G}} \left| w_g^{(r+1)} - w_g^{(r)} \right|,$$

Here, $\Delta_m^{(r)}$ and $\Delta_s^{(r)}$ are the relative changes in the natural-scale SI mean and SD, $\Delta_w^{(r)}$ is the largest absolute change among the four weights, and ϵ is a small positive constant that prevents division by zero. Numerical convergence is declared only when

$$\Delta_m^{(r)} < \text{tol}, \qquad \Delta_s^{(r)} < \text{tol}, \qquad \Delta_w^{(r)} < \text{tol}.$$

The symbol tol denotes the positive convergence tolerance, whose default value is 10^{-6}. Joint monitoring prevents termination based only on apparently stable SI parameters while pathway weights are still changing. The simulation analyses imposed a maximum of 300 iterations. Nevertheless, satisfying this stopping rule indicates numerical stability of the updates, not necessarily accurate recovery of the generating SI parameters, as demonstrated by the low-dispersion simulation and initialization sensitivity results.

7.2 Appendix B. Assessment of Component-Density Approximations

The Log-Normal extension uses computationally tractable working densities for the co-primary (CP), primary-tertiary (PT), and primary-quaternary (PQ) transmission pathways. As described in the Methods section, the ideal CP interval is the absolute difference of two independent Log-Normal serial intervals, whereas PT and PQ are sums of two and three independent intervals, respectively. The PS pathway comprises one direct serial interval and is represented exactly.

This appendix evaluates how closely these working densities approximate their ideal continuous distributions. The assessment concerns distributional approximation rather than bias in estimated SI parameters or the additional effects of rounding ICC intervals to calendar days.

7.2.1 B.1 Co-primary pathway: Half-Normal approximation

Let X_1 and X_2 denote independent serial intervals following the same Log-Normal distribution with natural-scale mean m and standard deviation s. The exact co-primary interval is $C_{\text{CP}} = |X_1 - X_2|$. Writing f_X for the density of one SI, its exact continuous density can be expressed as

$$f_{\text{CP}}^{\text{exact}}(c) = 2\int_0^\infty f_X(x) f_X(x+c)\,dx, \qquad c \geq 0.$$

The working model approximates the signed difference $X_1 - X_2$ by a Normal distribution with mean zero and variance $2s^2$, giving the Half-Normal density

$$f_{\text{CP}}^{\text{HN}}(c) = \frac{1}{s\sqrt{\pi}} \exp\left(-\frac{c^2}{4s^2}\right), \qquad c \geq 0.$$

This approximation preserves the variance scale of the signed difference before taking its absolute value, but does not necessarily reproduce the moments or shape of the exact CP distribution. In particular, the exact density depends on both m and s, whereas the Half-Normal approximation depends only on s. Differences in concentration near zero, curvature, and tail behaviour may therefore arise, especially when the underlying SI distribution is highly dispersed and right-skewed.

7.2.2 B.2 PT and PQ pathways: Distributional cost of the Fenton–Wilkinson approximation

The Fenton–Wilkinson (FW) approximation replaces a sum of independent Log-Normal serial intervals with a single Log-Normal distribution matching its exact natural-scale mean and variance.

For $S_n = \sum_{j=1}^{n} X_j$, the exact moments are $E(S_n) = nm$ and $\text{Var}(S_n) = ns^2$, where $n = 2$ for PT and $n = 3$ for PQ. Although the FW approximation preserves these two moments, it does not necessarily reproduce higher-order distributional characteristics, particularly skewness.

Let

$$c_v = \frac{s}{m}$$

denote the coefficient of variation of the underlying SI distribution.

The skewness of a single Log-Normal variable is $(c_v^2+3)c_v$. Since the serial intervals are independent, the skewness of their sum is

$$\gamma_{\text{sum},n} = \frac{(c_v^2+3)c_v}{\sqrt{n}}.$$

The moment-matched FW distribution has coefficient of variation $c_v/\sqrt{n}$. Its skewness is therefore

$$\gamma_{\text{FW},n} = \frac{(c_v^2/n+3)c_v}{\sqrt{n}}.$$

The resulting skewness difference is

$$\Delta\gamma_n = \gamma_{\text{sum},n} - \gamma_{\text{FW},n} = \frac{c_v^3(1-1/n)}{\sqrt{n}}, \qquad n > 1.$$

Because $\Delta\gamma_n > 0$, the FW approximation has lower overall positive skewness than the exact sum. The discrepancy increases with c_v^3, suggesting that the approximation becomes less accurate as the underlying SI distribution becomes more variable and strongly right-skewed.

Importantly, this result concerns the standardised third moment. It does not imply that the FW density is lower than the exact density at every point in the right tail.

7.2.3 B.3 Numerical assessment of approximation accuracy

To assess the magnitude of these distributional discrepancies, the ideal and working pathway-specific densities were compared under the three SI settings used in the main simulation: low dispersion $(m, s) = (15, 3)$, moderate dispersion $(2, 2)$, and high dispersion $(5, 10)$.

For CP, the exact absolute-difference density was evaluated numerically and compared with its Half-Normal approximation. For PT and PQ, exact sums of independent Log-Normal intervals were evaluated using numerical convolution and compared with their FW approximations.

The accuracy of each approximation was quantified using the density overlap coefficient,

$$\mathrm{OVL}(f,g) = \int_0^\infty \min\{f(c), g(c)\}\, dc = 1 - \frac{1}{2}\int_0^\infty |f(c) - g(c)|\, dc.$$

Values closer to 100% indicate greater agreement between the two continuous densities.

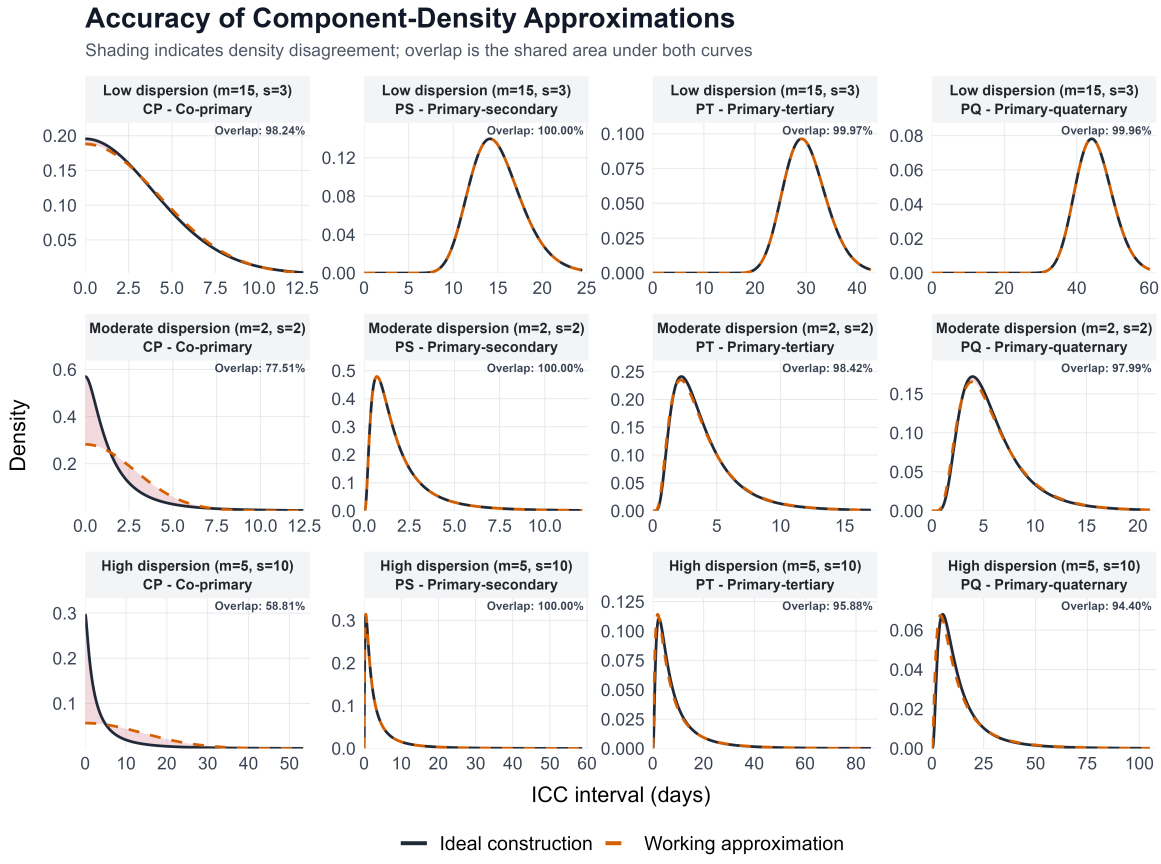


Figure 15: Comparison of ideal and working pathway-specific ICC densities under three Log-Normal SI dispersion settings. Rows represent low ($m = 15$, $s = 3$), moderate ($m = 2$, $s = 2$), and high ($m = 5$, $s = 10$) dispersion; columns correspond to CP, PS, PT, and PQ pathways. The horizontal axis shows ICC interval (days) and the vertical axis shows density. Solid dark curves represent the ideal continuous distributions, while dashed orange curves represent the working approximations; shaded regions indicate differences between them. CP uses a Half-Normal approximation to the absolute difference of two independent Log-Normal intervals, PS is exact, and PT/PQ use Fenton–Wilkinson approximations. Panel annotations report density overlap percentages calculated over the full numerical domain. Comparisons precede calendar-day discretisation. In the main simulation, CP observations were generated using the Half-Normal working model rather than the exact Log-Normal difference distribution.

As shown in Figure 15, the working densities closely approximate their ideal counterparts under low SI dispersion. The density overlaps are approximately 98.24% for CP, 99.97% for PT, and 99.96% for PQ, while the PS component is represented exactly.

As shown in Figure 15, the discrepancies become more pronounced as dispersion increases. Under moderate dispersion, CP overlap decreases to approximately 77.51%, compared with 98.42% and 97.99% for PT and PQ. Under high dispersion, CP overlap decreases further to approximately 58.81%, whereas PT and PQ retain overlaps of approximately 95.88% and 94.40%, respectively.

The numerical comparisons in Figure 15 are consistent with the theoretical expectations. The increasingly large FW skewness gap is reflected in declining density overlap under higher dispersion, although the overall PT/PQ approximations remain relatively close. In contrast, the Half-Normal approximation can differ substantially from the exact CP density, particularly near zero.

The results in Figure 15 indicate that approximation accuracy depends on the underlying SI dispersion and transmission pathway. However, density overlap does not directly quantify bias in EM-based SI estimation. The effect of approximation error on estimated parameters also depends on pathway weights, discretised likelihood contributions, and posterior pathway responsibilities.

Furthermore, because the main simulation generated CP observations using the Half-Normal working model, it did not separately assess the additional parameter-estimation error that might arise when real CP intervals follow the exact absolute difference of independent Log-Normal serial intervals.

7.3 Supplementary simulation results

The following figures provide the complete simulation summaries for sample sizes other than the $N = 100$ reference setting. Each figure displays the distribution of parameter-estimation bias for converged runs and the convergence rate over all 100 repetitions. The iteration cap is 300. The RMSE summaries for all 36 simulation configurations are given in Table 7.

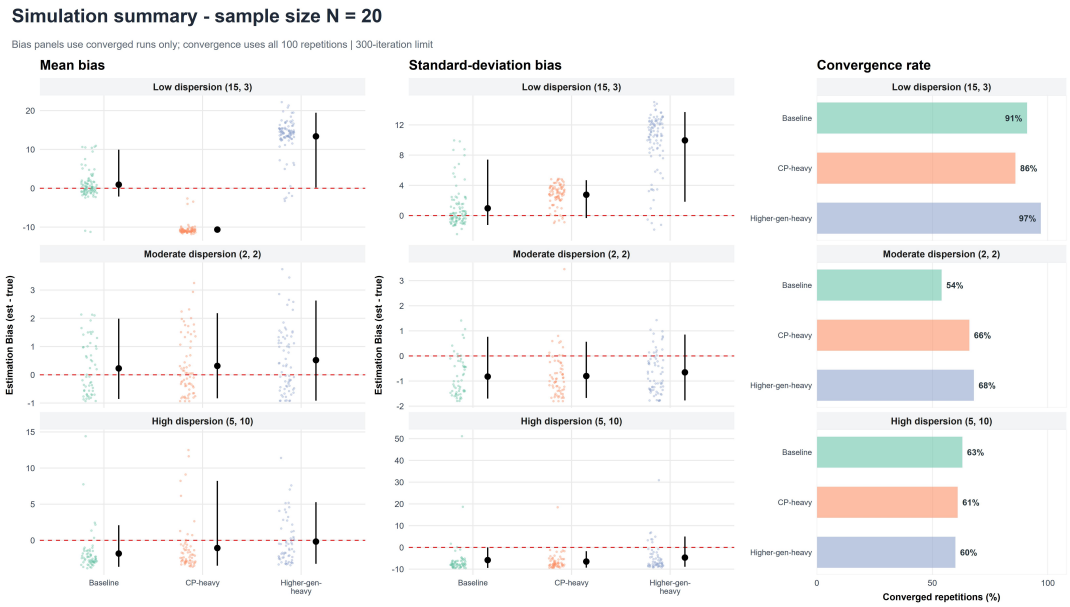


Figure 16: Simulation summary for N = 20 across nine scenarios. Rows show low, moderate, and high SI dispersion; the horizontal categories are baseline, CP-heavy, and higher-generation-heavy pathway profiles. Left and middle panels show estimated minus true SI mean and SD (days) for converged runs: coloured dots are repetitions, black points are mean biases with 5th–95th percentile ranges, and red dashed lines mark zero. Right panels show convergence percentage across all 100 repetitions, with profile vertically and percentage horizontally. Compare recovery and convergence under the 300-iteration cap.

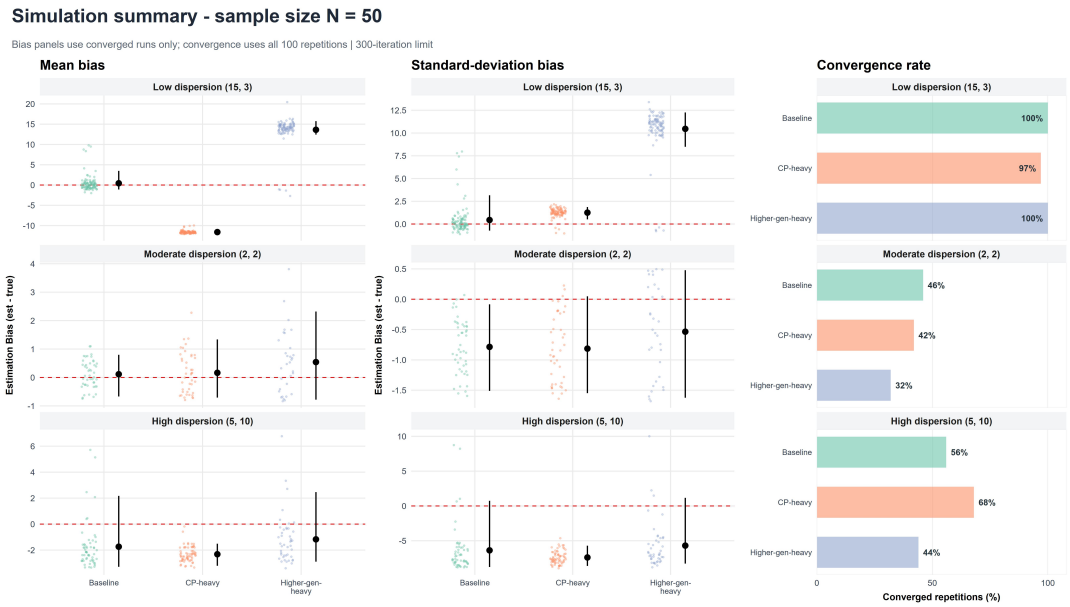


Figure 17: Simulation summary for N = 50 across nine scenarios. Rows show low, moderate, and high SI dispersion; the horizontal categories are baseline, CP-heavy, and higher-generation-heavy pathway profiles. Left and middle panels show estimated minus true SI mean and SD (days) for converged runs: coloured dots are repetitions, black points are mean biases with 5th–95th percentile ranges, and red dashed lines mark zero. Right panels show convergence percentage across all 100 repetitions, with profile vertically and percentage horizontally. Compare recovery and convergence under the 300-iteration cap.

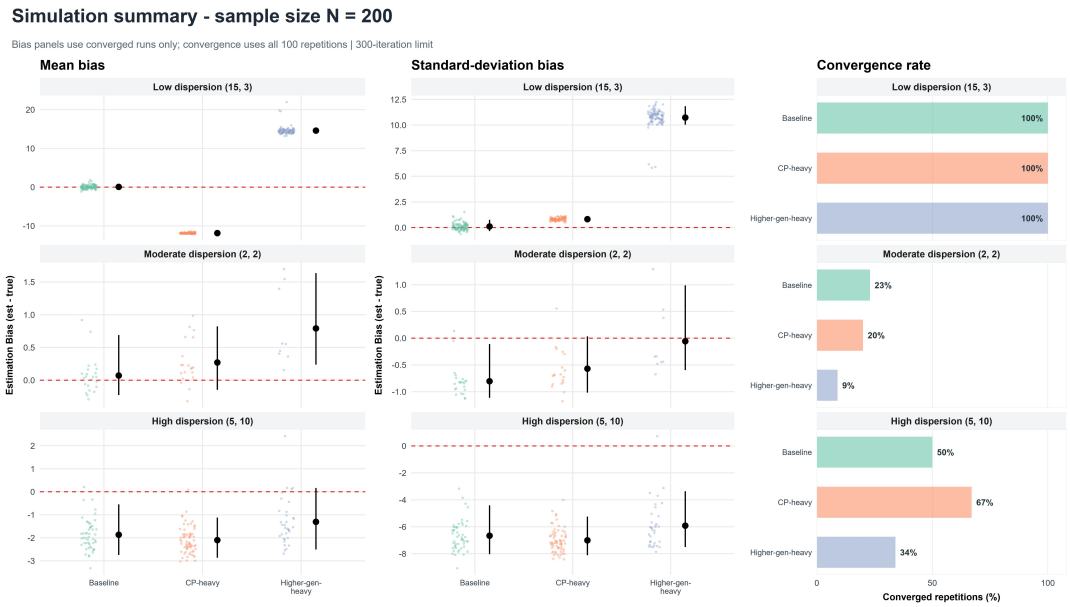


Figure 18: Simulation summary for N = 200 across nine scenarios. Rows show low, moderate, and high SI dispersion; the horizontal categories are baseline, CP-heavy, and higher-generation-heavy pathway profiles. Left and middle panels show estimated minus true SI mean and SD (days) for converged runs: coloured dots are repetitions, black points are mean biases with 5th–95th percentile ranges, and red dashed lines mark zero. Right panels show convergence percentage across all 100 repetitions, with profile vertically and percentage horizontally. Compare recovery and convergence under the 300-iteration cap.

Table 7: RMSE and convergence for all 36 combinations of sample size, SI dispersion, and transmission-pathway profile. RMSE uses finite converged estimates; convergence uses all repetitions.

N	Dispersion	Profile	Mean RMSE	SD RMSE	Conv. (%)
20	Low (15, 3)	Baseline	3.71	2.78	91
20	Low (15, 3)	CP-heavy	10.74	3.11	86
20	Low (15, 3)	Higher-generation-heavy	14.20	10.63	97
20	Moderate (2, 2)	Baseline	0.98	1.14	54
20	Moderate (2, 2)	CP-heavy	1.09	1.17	66
20	Moderate (2, 2)	Higher-generation-heavy	1.26	1.05	68
20	High (5, 10)	Baseline	3.28	10.09	63
20	High (5, 10)	CP-heavy	3.64	7.57	61
20	High (5, 10)	Higher-generation-heavy	3.16	7.81	60
50	Low (15, 3)	Baseline	2.07	1.68	100
50	Low (15, 3)	CP-heavy	11.60	1.36	97
50	Low (15, 3)	Higher-generation-heavy	14.01	10.75	100
50	Moderate (2, 2)	Baseline	0.50	0.90	46

Table 7: RMSE and convergence for all 36 combinations of sample size, SI dispersion, and transmission-pathway profile. RMSE uses finite converged estimates; convergence uses all repetitions.

N	Dispersion	Profile	Mean RMSE	SD RMSE	Conv. (%)
50	Moderate (2, 2)	CP-heavy	0.74	0.99	42
50	Moderate (2, 2)	Higher-generation-heavy	1.22	0.91	32
50	High (5, 10)	Baseline	2.51	7.27	56
50	High (5, 10)	CP-heavy	2.39	7.45	68
50	High (5, 10)	Higher-generation-heavy	2.20	6.66	44
100	Low (15, 3)	Baseline	0.66	0.43	100
100	Low (15, 3)	CP-heavy	11.73	0.99	100
100	Low (15, 3)	Higher-generation-heavy	14.49	10.96	100
100	Moderate (2, 2)	Baseline	0.50	0.93	37
100	Moderate (2, 2)	CP-heavy	0.60	0.84	32
100	Moderate (2, 2)	Higher-generation-heavy	1.05	0.84	24
100	High (5, 10)	Baseline	2.16	6.81	48
100	High (5, 10)	CP-heavy	2.27	7.27	53
100	High (5, 10)	Higher-generation-heavy	1.83	6.63	37
200	Low (15, 3)	Baseline	0.49	0.36	100
200	Low (15, 3)	CP-heavy	11.86	0.83	100
200	Low (15, 3)	Higher-generation-heavy	14.63	10.77	100
200	Moderate (2, 2)	Baseline	0.28	0.86	23
200	Moderate (2, 2)	CP-heavy	0.43	0.69	20
200	Moderate (2, 2)	Higher-generation-heavy	0.96	0.62	9
200	High (5, 10)	Baseline	1.99	6.76	50
200	High (5, 10)	CP-heavy	2.18	7.06	67
200	High (5, 10)	Higher-generation-heavy	1.66	6.16	34

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